Smart Hospital Management System (SmartCare)

Consolidated PBL Reports — Daily & Weekly Documentation

Team Nobita (PBL2627-AI&DS-A-031;) | Mentor: Er. Ram Babu Buri
Java PBL Academic Year 2026–27 | Period: 06-07-2026 to 04-10-2026 (13 Weeks, 65 Working Days)

No.	Member Name & Roll No.	Module Assigned	Total Logged Hours
1	Devansh Mathur (24E1ARADM40P040)	Module 1 – Authentication, Admin and Security	115.75 hrs
2	Deepak Kumar (24E1ARADM30P035)	Module 2 – Patient and Doctor Management	113.25 hrs
3	Hansraj Pandey (24E1ARADM40P055)	Module 3 – Appointment and Consultation	116.25 hrs
4	Aryan Daware (24E1ARADM30P020)	Module 4 – Pharmacy, Laboratory and Nursing	112.00 hrs
5	Devkinandan Kumawat (24E1ARADM30P042)	Module 5 – Billing, Admission, Reports and Feedback	116.75 hrs
TOTAL	Team Nobita (5 Members)	Complete System (5 Integrated Modules)	574.00 hrs

Document Purpose & Portal Submission Instructions

This official consolidated document compiles the verified weekly status reports and day-wise technical activity logs for all five members of Team Nobita. Each Weekly Report contains an in-depth synthesis of approximately 80 words detailing key milestones achieved during that sprint. Each Daily Work Entry features an expanded technical description strictly between 50 and 60 words, articulating implementation specifics, architectural decisions, and verification results across 65 working days (Monday to Friday, 1.5 to 2.0 hours per entry).

Submission Guide for PBL Portal:
• Weekly Report Tab: Select corresponding Week, insert the central GitHub repository link, and paste the three respective fields (Work Completed, Problems Faced, and Plan for Next Week).
• Daily Work Log Tab: Select target Date, input Hours Worked, retain Status as Completed, and copy-paste the specific day's 50–60 word technical work description.
• GitHub Repository: https://github.com/mathurdevansh/Smart-Hospital-Management-System

1. Devansh Mathur (24E1ARADM40P040)

Module 1 - Authentication, Admin and Security

Member	Devansh Mathur (24E1ARADM40P040)
Module	Module 1 - Authentication, Admin and Security
Code owned	LoginServlet, LogoutServlet, AuthService, PasswordUtil (BCrypt), AuthenticationFilter, AuthorizationFilter, AdminServlet, UserDAO, DepartmentDAO, AuditDAO, DBConnection, admin JSPs (dashboard, users, departments, audit), Docker + Render + cloud MySQL deployment.
Period	06-07-2026 to 04-10-2026 (13 weeks). Daily hours between 1.5 and 2.0.
GitHub link	https://github.com/mathurdevansh/Smart-Hospital-Management-System

Part A - Weekly Reports (13 Weeks) — Devansh Mathur

Week 1 | 06-07-2026 to 12-07-2026 | Phase: Team formation, guide selection and abstract

Week dropdown	Week 1
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Convened with the team to finalize the Smart Hospital Management System project scope and architecture. Conducted domain research into administrative challenges within manual hospital record keeping, detailing core authentication and role-based access requirements. Authored the comprehensive module abstract focusing on user management, security filters, and administrative control. Consulted with our faculty mentor to validate the Servlet-JSP and MySQL technology stack. Collaborated with teammates to integrate module components, refine the system description, and successfully submit the unified abstract to the portal.
Problems Faced	Difficulty in choosing one idea among several. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Identified five user roles: Admin, Doctor, Patient, Nurse and Receptionist. (2) Wrote functional requirements for login, logout, session handling and role-based access. (3) Wrote security requirements: password hashing, session timeout, audit trail.

Week 2 | 13-07-2026 to 19-07-2026 | Phase: SRS (Software Requirement Specification)

Week dropdown	Week 2
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Formulated the Software Requirement Specification for Module 1, establishing authentication, user management, and security boundaries. Outlined explicit functional criteria for user onboarding, credential verification, session lifecycle management, and role-based authorization across five key actors. Formulated strict enterprise security requirements including BCrypt password encryption, idle session termination, and audit log generation. Documented non-functional standards addressing database throughput, high availability, and interface usability. Consolidated all security specifications into the formal team SRS document and conducted an architectural walkthrough with teammates to verify functional completeness.
Problems Faced	Unclear which roles needed separate dashboards. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Drew the use-case diagram for the Admin actor (manage users, departments, audit logs). (2) Drew the login and logout sequence diagram with filters and session. (3) Designed class diagram for User, Role and AuditLog.

Week 3 | 20-07-2026 to 26-07-2026 | Phase: UML design (use-case, class, sequence)

Week dropdown	Week 3
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Engineered comprehensive Unified Modeling Language diagrams for the authentication and administrative subsystems. Drafted high-level use-case diagrams illustrating administrative privileges including user account management, departmental allocation, and audit tracking. Formulated sequence diagrams capturing dynamic interactions across authentication filters, HTTP session managers, and credential repositories during login and logout. Designed detailed class diagrams for User, Role, and AuditLog domain entities with appropriate multiplicity and relationships. Merged all module diagrams into the centralized team design specification following mentor feedback and architectural guidelines for project implementation.
Problems Faced	Mixing class diagram relations for User and Role. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Designed ER tables roles, users and audit_logs with keys and constraints. (2) Designed the departments table and linked doctors to departments. (3) Created Maven project structure with pom.xml and Jakarta Servlet dependencies.

Week 4 | 27-07-2026 to 02-08-2026 | Phase: Database (ER) design, UI mock-ups, project setup

Week dropdown	Week 4
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Designed relational database schemas and set up foundational development infrastructure for the hospital management platform. Created normalized Entity-Relationship tables for user accounts, assigned roles, and audit trail logs incorporating strict primary and foreign key constraints. Structured the department table ensuring seamless relational linkage with medical staff. Configured the multi-module Maven project directory with Jakarta Servlet 6.0 and MySQL connector dependencies. Established the team GitHub repository, published structural conventions, and delivered interface mock-ups for administrative dashboards, user management screens, and system security controls.
Problems Faced	Foreign key order created errors in SQL script. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote the Role and User model classes with getters and setters. (2) Created DBConnection utility reading db.properties. (3) Implemented UserDAO methods to fetch user by username and email.

Week 5 | 03-08-2026 to 09-08-2026 | Phase: Module coding - models and DAO layer

Week dropdown	Week 5
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Implemented foundational data access layer components and domain models for authentication and administration. Authored robust User and Role Java model classes with complete encapsulated fields, constructors, and property accessors. Developed a centralized database connection manager utilizing DBCP connection pooling and externalized property files. Constructed UserDAO implementing secure PreparedStatement queries for user retrieval, registration, updates, and profile deactivations. Built DepartmentDAO supporting complete CRUD operations and verified relational integrity using unit tests against the local MySQL instance to confirm error-free transaction execution.
Problems Faced	Driver not found error for MySQL connector. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote PasswordUtil using the jBCrypt library for hashing and verification. (2) Built AuthService login flow with credential and role checking. (3) Implemented LoginServlet doGet and doPost with redirect by role.

Week 6 | 10-08-2026 to 16-08-2026 | Phase: Module coding - services and servlets

Week dropdown	Week 6
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Developed core application business services and HTTP servlet controllers handling system authentication and session management. Implemented PasswordUtil utilizing the jBCrypt library for cryptographic salt generation and hash verification. Built AuthService to authenticate user credentials, check account active status, and determine role privileges. Programmed LoginServlet to manage incoming GET and POST requests with dynamic redirection based on user roles. Created LogoutServlet to terminate sessions cleanly, and integrated AuditDAO to capture login timestamps and administrative actions with precision across all hospital user categories.
Problems Faced	Role redirect looped back to login page. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Created login.jsp with responsive form and error messages. (2) Created admin dashboard.jsp with summary cards. (3) Built admin users.jsp with table for add, edit and delete users.

Week 7 | 17-08-2026 to 23-08-2026 | Phase: Module coding - JSP views and UI

Week dropdown	Week 7
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Designed and developed responsive JavaServer Pages for administrative functions and authentication workflows. Created login.jsp with client-side form controls, visual alert banners, and customized styling. Built the administrative dashboard interface featuring quick-access statistics cards, department summaries, and recent activity monitors. Developed admin user management interfaces supporting real-time staff registration, record editing, and role assignment tables. Constructed views for departments and system audit logs, establishing modular layouts through reusable navigation bars, headers, sidebars, and stylesheet includes to ensure visual consistency.
Problems Faced	Include files broke relative paths in admin pages. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote ValidationUtil for email, phone and empty field checks. (2) Created ApplicationException and DatabaseException classes. (3) Implemented AuthenticationFilter to block pages for logged-out users.

Week 8 | 24-08-2026 to 30-08-2026 | Phase: Module coding - validation, exceptions, security

Week dropdown	Week 8
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Implemented comprehensive application validation, centralized exception handling, and security filter chains. Programmed ValidationUtil containing robust regular expression checkers for email addresses, phone formats, and mandatory form fields. Created custom ApplicationException and DatabaseException classes to standardize error handling across service and DAO layers. Configured AuthenticationFilter to prevent unauthorized access to protected paths by unauthenticated users. Developed AuthorizationFilter enforcing strict role-based access control, and configured session timeouts, secure cookie flags, and HTTP error pages in web.xml for production security compliance.
Problems Faced	Filter blocked static CSS and JS files. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Completed AdminServlet actions for user and department management. (2) Added duplicate username and email validation in user creation. (3) Added activate and deactivate user option for admin.

Week 9 | 31-08-2026 to 06-09-2026 | Phase: Module coding - module completion and self-review

Week dropdown	Week 9
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Completed administrative servlet controllers and conducted exhaustive code optimization prior to team integration. Finalized AdminServlet actions allowing administrators to create, update, search, and toggle active statuses of hospital staff accounts. Implemented server-side validation preventing duplicate username and email registrations across user roles. Refactored DAO and service class methods by removing redundant database queries and inserting comprehensive Javadoc comments. Successfully pushed polished module code to the collaborative GitHub repository and updated tracking matrices for integration to prepare for multi-module assembly.
Problems Faced	Merge conflicts in shared files. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Merged all modules into one WAR project and resolved package conflicts. (2) Connected the login redirect to every role dashboard. (3) Checked filter rules against every servlet URL pattern.

Week 10 | 07-09-2026 to 13-09-2026 | Phase: Integration of all five modules

Week dropdown	Week 10
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Coordinated the integration of all five independent modules into a unified hospital enterprise web archive. Resolved Maven dependency conflicts, package naming collisions, and shared resource overlaps across application layers. Configured centralized authentication redirection directing administrators, doctors, receptionists, nurses, and patients to their respective dashboards post-login. Harmonized global navigation bars, headers, and access control filter mappings across all application servlet endpoints. Executed comprehensive end-to-end integration runs verifying smooth user transitions from login through clinical workflows to logout across all system modules.
Problems Faced	Sidebar links showed wrong items for some roles. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Prepared test cases for login, wrong password and locked access. (2) Tested role-based access by opening restricted URLs directly. (3) Tested session expiry and logout behaviour.

Week 11 | 14-09-2026 to 20-09-2026 | Phase: Testing (unit, functional, role-based)

Week dropdown	Week 11
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Executed rigorous unit, functional, and security test suites across integrated system endpoints. Formulated test plans covering positive and negative authentication scenarios including incorrect passwords, SQL injection attempts, and locked accounts. Tested AuthorizationFilter boundaries by attempting direct unauthorized URL manipulation across distinct user roles. Simulated concurrent sessions, idle timeouts, and credential invalidations to confirm session isolation. Identified layout anomalies, validation oversights, and role redirect defects, collaborating with teammates to implement immediate code fixes and verify resolution through regression test cycles.
Problems Faced	Session lost after server restart during tests. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote a Dockerfile (Maven build stage and Tomcat 10.1 run stage). (2) Created a cloud MySQL instance and moved the schema to it. (3) Updated db.properties for the cloud database with SSL.

Week 12 | 21-09-2026 to 27-09-2026 | Phase: Deployment (Docker, Render, cloud MySQL) and bug fixing

Week dropdown	Week 12
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Engineered deployment automation pipelines and deployed the hospital application to cloud infrastructure. Authored a multi-stage Dockerfile utilizing Maven for compiling the WAR package and Apache Tomcat 10.1 for runtime execution. Provisioned a managed cloud MySQL database instance, migrated schema definitions, and populated baseline administrative seed records. Configured production environment variables with SSL database connection strings and deployed containerized builds to Render. Conducted extensive smoke testing on the live environment and resolved minor session cookie issues across multiple browser client sessions.
Problems Faced	Cloud DB SSL connection refused at first. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote the introduction and tools sections of the project report. (2) Documented authentication and security design in the report. (3) Prepared slides on architecture, roles and security.

Week 13 | 28-09-2026 to 04-10-2026 | Phase: Report, PPT, demo video and viva preparation

Week dropdown	Week 13
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Compiled final project documentation, recorded system demonstrations, and completed viva presentation preparation. Authored comprehensive report chapters covering system architecture, security implementation, BCrypt cryptography, and role-based filtering mechanisms. Designed professional presentation slides illustrating application architecture, data flow diagrams, and administrative capabilities. Recorded and edited an end-to-end video demonstration showcasing administrative controls, user management, and security workflows. Conducted technical defense mock sessions, reviewing filter mechanics, session management protocols, and database connection pooling to prepare for viva evaluations with our mentor.
Problems Faced	Video recording took multiple attempts. Solved with team discussion and guide's suggestions.
Plan for Next Week	Final viva and demo, submit report, PPT, SRS, demo video and working software on the portal.

Part B - Daily Work Log (65 Entries) — Devansh Mathur

Status for all entries: Completed. Saturday is the weekly mentor meeting, so no daily entry is listed.

Week 1 - Team formation, guide selection and abstract

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
06-07-2026 (Mon)	2	Completed	Collaborated with all five team members during our initial project kickoff meeting to brainstorm prospective domain solutions. We systematically evaluated operational bottlenecks in community healthcare facilities and unanimously shortlisted the Smart Hospital Management System. We outlined the preliminary project vision, established team communication channels, and demarcated individual module ownership across the proposed five enterprise subsystems.
07-07-2026 (Tue)	1.5	Completed	Conducted an in-depth analytical study investigating operational vulnerabilities in conventional hospital record-keeping methodologies. I documented systemic issues surrounding physical paper registries, misplaced diagnostic files, and uncoordinated role handoffs. Based on these findings, I drafted foundational system specifications for unified user authentication, role hierarchies, credential protection standards, and role-based administrative control mechanisms across the institution.
08-07-2026 (Wed)	2	Completed	Authored the formal abstract section specifically addressing authentication, administrative control, and security infrastructure for our hospital platform. I detailed the technical necessity of robust role segregation between administrators, medical doctors, desk receptionists, clinical nurses, and patients. Furthermore, I highlighted how structured administrative governance eliminates unauthorized data modifications and preserves critical hospital clinical records.
09-07-2026 (Thu)	1.75	Completed	Met with our faculty project guide, Er. Ram Babu Buri, to evaluate proposed technology architectures against department research standards. We analyzed architectural trade-offs between frameworks and confirmed an enterprise Java Servlet and JavaServer Pages stack paired with MySQL relational storage. We established coding guidelines, version control standards, and verified project scope feasibility under academic timeline constraints.
10-07-2026 (Fri)	2	Completed	Conducted a comprehensive review and proofreading session of the finalized project abstract alongside all five team members. I verified that authentication, role definitions, and administrative security boundaries aligned with peer modules. After incorporating constructive feedback from our guide, I formatted the final document and assisted the team with its formal submission through the portal.

Week 2 - SRS (Software Requirement Specification)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
13-07-2026 (Mon)	1.75	Completed	Formulated the definitive user persona taxonomy for the hospital management system, identifying five distinct actor roles: Administrator, Doctor, Patient, Nurse, and Receptionist. I outlined specific operational responsibilities and system interaction boundaries for each role, ensuring that privilege separation prevents privilege escalation and guarantees that users only interact with authorized hospital interface modules.
14-07-2026 (Tue)	1.75	Completed	Drafted comprehensive functional requirements governing user authentication workflows, credential verification, session management, and role-based route protection. I specified procedural state flows for account login, graceful logout, idle session invalidation, and automated redirection based on user privilege tiers. Additionally, I documented explicit error handling behaviors for invalid credentials, deactivated profiles, and unauthorized endpoint requests.
15-07-2026 (Wed)	1.5	Completed	Formulated stringent enterprise security specifications to protect hospital records and safeguard administrative operations. I mandated BCrypt one-way password hashing with per-user cryptographic salts to prevent rainbow table attacks. Furthermore, I specified a thirty-minute idle session timeout, HTTP-only secure cookie configurations, and comprehensive audit trail logging capturing every administrative user state change.
16-07-2026 (Thu)	1.75	Completed	Authored rigorous non-functional requirements within the Software Requirement Specification, focusing on system performance, operational availability, and UI accessibility. I established quantitative response benchmarks requiring authentication roundtrips under two hundred milliseconds. Additionally, I documented strict uptime criteria, cross-browser compatibility standards, responsive viewport parameters, and localized data validation error presentation across all administrative interfaces.
17-07-2026 (Fri)	1.5	Completed	Consolidated the authentication, administrative governance, and security requirement sections into the master Software Requirement Specification document. I led an internal review walkthrough with all team members to confirm that login handoffs and authorization boundaries aligned with medical records and billing workflows. We polished the technical language and formally cataloged the baseline requirements.

Week 3 - UML design (use-case, class, sequence)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
20-07-2026 (Mon)	1.5	Completed	Engineered the formal Unified Modeling Language use-case diagram for the Administrator actor utilizing Visual Paradigm. I mapped out comprehensive administrative functional boundaries, including creating new staff profiles, modifying department assignments, inspecting real-time audit logs, and toggling account statuses. I modeled include and extend relationships to represent credential validation dependencies and session logging mechanics accurately.
21-07-2026 (Tue)	2	Completed	Constructed detailed sequence diagrams illustrating the dynamic runtime interactions during user login and logout lifecycles. I modeled message exchanges flowing across web browser clients, servlet container filters, the central authentication servlet, business service layers, and the database DAO. The diagram clearly depicts credential verification, session token generation, security context establishment, and role-based dashboard redirection.
22-07-2026 (Wed)	1.75	Completed	Designed the structural class diagram for core security and domain entities, specifically detailing the User, Role, and AuditLog classes. I defined private instance attributes, public getters and setters, constructor signatures, and critical operational methods. Furthermore, I established relational associations and multiplicity constraints connecting user accounts with role permissions and tracking actions in audit logs.
23-07-2026 (Thu)	1.5	Completed	Developed the domain class diagram modeling the hospital Department entity and its structural relationship with medical staff profiles. I specified attributes including department identifier, department name, facility location, and description. I established a one-to-many association linking each department to assigned doctors, ensuring that medical consultations and staff scheduling maintain clean relational references across modules.
24-07-2026 (Fri)	2	Completed	Collaborated with module owners to merge our authentication, user governance, and security UML diagrams into the team design specification. We resolved structural inconsistencies between doctor account entities and patient profile models, standardizing naming conventions and datatype definitions across class diagrams. I updated the consolidated UML repository and confirmed alignment with our guide's technical guidelines.

Week 4 - Database (ER) design, UI mock-ups, project setup

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
27-07-2026 (Mon)	1.5	Completed	Architected relational database tables for roles, users, and audit_logs in MySQL, ensuring strict normalization to Third Normal Form. I defined primary key constraints, unique indexes on usernames and email addresses, foreign key relationships, and created_at timestamps. These schema definitions prevent orphan records, enforce enterprise data integrity, and guarantee efficient indexing during authentication lookup queries.
28-07-2026 (Tue)	1.75	Completed	Designed the departments relational table schema and engineered its foreign key associations connecting doctors and administrative personnel. I structured column definitions for department identification, departmental titles, physical building locations, and status flags. Additionally, I added relational integrity constraints ensuring that deleting or updating department records will not leave dangling doctor assignments in secondary tables.
29-07-2026 (Wed)	1.75	Completed	Initialized the enterprise Maven project structure and configured pom.xml with essential web application dependencies. I incorporated Jakarta Servlet API 6.0, Jakarta Server Pages, JSTL library implementations, and MySQL Connector-J drivers. Furthermore, I configured the Maven compiler plugin for Java 17 compatibility and structured the standard directory layout for source code, webapp resources, and test suites.
30-07-2026 (Thu)	1.75	Completed	Created the official team GitHub repository and established branching workflows, pull request conventions, and commit guidelines. I published the baseline Maven directory skeleton, created a comprehensive gitignore file preventing IDE and compiled artifact commits, and distributed repository access to all five team members. We verified local repository cloning and established feature-branch workflows.
31-07-2026 (Fri)	1.5	Completed	Crafted high-fidelity user interface wireframes and mock-ups for the primary authentication portal and administrator dashboard. Using modern layout tools, I designed responsive login forms with error state visualizations, password visibility toggles, and clean branding elements. For the administrative dashboard, I mapped out statistical overview metric cards, department listing tables, and recent audit activity streams.

Week 5 - Module coding - models and DAO layer

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
03-08-2026 (Mon)	1.5	Completed	Programmed the User and Role domain model classes in Java adhering strictly to standard JavaBean encapsulation principles. I defined private fields for user identification, username, hashed password, email, role identifier, and active status flag. I implemented complete parameter constructors, no-argument constructors, accessors, mutators, and overridden toString, equals, and hashCode methods to support collection operations.
04-08-2026 (Tue)	1.5	Completed	Engineered the centralized DBConnection utility class to establish robust, performant JDBC connections with the MySQL database. I integrated Apache Commons DBCP connection pooling to eliminate the overhead of repeatedly creating physical connections. The utility reads database credentials, driver class names, and connection pool configurations securely from an externalized db.properties file in the classpath.
05-08-2026 (Wed)	2	Completed	Implemented the UserDAO interface and developed core query methods to retrieve user records by unique username and email address. Using JDBC PreparedStatement implementations, I parameterized input values to eliminate SQL injection vulnerabilities completely. I mapped database query ResultSet rows directly into instantiated User objects, handling connection closures safely within modern Java try-with-resources structural blocks.
06-08-2026 (Thu)	1.75	Completed	Expanded UserDAO by implementing comprehensive data persistence methods for inserting new staff accounts, updating contact details, and toggling user activation flags. I ensured that all transactional operations verify row update counts and encapsulate underlying SQLExceptions into domain-specific exceptions. I tested record creation directly against our local MySQL test database to verify constraint compliance.
07-08-2026 (Fri)	1.75	Completed	Constructed the DepartmentDAO class providing full Create, Read, Update, and Delete capabilities for hospital departments. I authored parameterized SQL statements to fetch all active departments, locate departments by unique identifier, and save new hospital divisions. I executed rigorous unit tests against our local MySQL schema, confirming reliable data retrieval and foreign key integrity.

Week 6 - Module coding - services and servlets

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
10-08-2026 (Mon)	2	Completed	Constructed the PasswordUtil cryptographic helper class leveraging the jBCrypt library for industry-standard password security. I implemented static utility methods to generate cryptographically random salts and hash plaintext passwords using the blowfish algorithm. Additionally, I built verification logic that compares raw candidate passwords against stored hashes, successfully preventing timing attacks and protecting user credentials.
11-08-2026 (Tue)	1.75	Completed	Developed the AuthService business logic layer to orchestrate the complete credential authentication and authorization lifecycle. I programmed login verification logic that queries UserDAO, checks account active flags, and verifies password hashes through PasswordUtil. Furthermore, I integrated role retrieval mechanisms to associate valid credentials with specific permission profiles required for role-based dashboard redirection.
12-08-2026 (Wed)	1.5	Completed	Engineered the LoginServlet controller to process incoming HTTP GET and POST authentication requests. In the doGet handler, the servlet forwards unauthenticated requests to the login JSP form. In the doPost handler, it extracts user credentials, invokes AuthService, establishes an authenticated HTTP session upon success, and redirects users dynamically to their role-specific dashboard endpoints.
13-08-2026 (Thu)	1.75	Completed	Implemented the LogoutServlet to facilitate secure, clean user session termination across all hospital application modules. The servlet retrieves the active HttpSession, invalidates session attributes, unbinds authenticated principal tokens, and clears browser session cookies. It ensures that post-logout browser back-button navigation cannot access cached authenticated pages, redirecting users to the login screen with a confirmation message.
14-08-2026 (Fri)	1.5	Completed	Created the AuditDAO class to record and track critical security events and administrative actions across the hospital platform. I programmed parameterized insert queries that capture event timestamps, acting user identifiers, source IP addresses, and descriptive event types like successful logins, failed attempts, and profile modifications. I validated automated audit log insertions during authentication operations.

Week 7 - Module coding - JSP views and UI

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
17-08-2026 (Mon)	2	Completed	Developed the user-facing login.jsp view utilizing clean semantic HTML5 markup and modern responsive CSS styling. I designed intuitive input form fields for username and password, complemented by interactive validation indicators. Furthermore, I implemented JSTL conditional tags to dynamically render dismissible error alerts whenever invalid credentials, expired sessions, or unauthorized access attempts are passed from servlets.
18-08-2026 (Tue)	1.5	Completed	Constructed the responsive admin dashboard.jsp view providing hospital administrators with a unified control cockpit. I designed modular summary metric cards showcasing total active staff, registered patients, scheduled appointments, and departmental counts. I incorporated quick-action navigation buttons and ensured the layout adapts gracefully across desktop, tablet, and mobile viewport resolutions using CSS flexbox grids.
19-08-2026 (Wed)	2	Completed	Built the admin users.jsp interface to facilitate complete staff account management and administrative oversight. I implemented a responsive tabular view listing staff members with dynamic columns for username, email, assigned role, and status. I integrated modal dialog forms enabling administrators to register new users, update existing employee records, and deactivate compromised user accounts effortlessly.
20-08-2026 (Thu)	2	Completed	Developed the admin departments.jsp and audit.jsp interfaces to deliver comprehensive institutional monitoring. The departments view provides administrators with intuitive tools to create and edit hospital wings, while the audit view presents a chronologically sorted activity log with pagination. I added status indicators and timestamp formatters to ensure administrative clarity during security reviews.
21-08-2026 (Fri)	2	Completed	Architected reusable JSP layout templates including common header.jsp, navbar.jsp, sidebar.jsp, and footer.jsp partials. I centralized shared styling by linking style.css across all views and modularized navigation links based on distinct user roles. I resolved relative path resolution challenges using context-aware URL expressions, guaranteeing seamless template reusability, modular layout consistency, and maintainability across all modules.

Week 8 - Module coding - validation, exceptions, security

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
24-08-2026 (Mon)	1.75	Completed	Engineered the ValidationUtil utility class containing comprehensive server-side input verification methods for all web forms. I implemented rigorous regular expression patterns to validate standard email formats, ten-digit phone numbers, and non-empty string fields. Furthermore, I included boundary checks ensuring numeric parameters remain within valid ranges, preventing corrupt data from propagating into service layers.
25-08-2026 (Tue)	1.5	Completed	Architected custom application exception classes, specifically ApplicationException and DatabaseException, to establish a unified hierarchy for error handling across the platform. I refactored service and DAO layer methods to catch low-level SQLExceptions, wrap them into descriptive unchecked domain exceptions, and preserve original root-cause stack traces for streamlined administrative debugging and clean error reporting.
26-08-2026 (Wed)	2	Completed	Programmed the AuthenticationFilter servlet filter to intercept all incoming HTTP requests directed toward protected hospital URL patterns. The filter inspects active session contexts to verify whether an authenticated user principal exists. If an unauthenticated user attempts to access internal resources directly, the filter halts execution and redirects them securely to the login portal.
27-08-2026 (Thu)	2	Completed	Implemented the AuthorizationFilter to establish strict role-based access control across all application endpoints. The filter evaluates the authenticated user's role against declared access matrices for administrative, medical, nursing, reception, and patient paths. If a user attempts to access an unauthorized endpoint, the filter intercepts the request, returning an HTTP 403 Forbidden access denial view.
28-08-2026 (Fri)	1.75	Completed	Configured critical web application deployment descriptors within web.xml to harden overall system security. I established a thirty-minute session timeout threshold, configured HttpOnly and SameSite cookie flags to mitigate cross-site scripting risks, and mapped standard HTTP error status codes 403, 404, and 500 to custom, user-friendly error display pages throughout the application.

Week 9 - Module coding - module completion and self-review

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
31-08-2026 (Mon)	1.5	Completed	Finalized all controller actions within AdminServlet to support full administrative lifecycle management of hospital entities. I implemented robust doGet and doPost dispatch routines handling user creation, profile updates, account deactivations, and departmental reorganizations. I verified that all incoming administrative requests validate session authorization before executing sensitive modifications against backend data access objects.
01-09-2026 (Tue)	2	Completed	Implemented dual-layer uniqueness validation within AdminServlet to prevent duplicate username and email registrations. Prior to persisting new hospital staff records, the servlet queries UserDAO to verify whether candidate credentials already exist in the database. If collisions occur, the servlet halts processing and returns localized, actionable error messages directly onto the registration form.
02-09-2026 (Wed)	2	Completed	Engineered user status toggle functionality allowing administrators to activate or deactivate hospital staff accounts dynamically. I added a dedicated action handler in AdminServlet that modifies the active boolean flag in the database without destroying historical user data. This ensures former employees are immediately barred from authenticating while preserving all associated audit logs.
03-09-2026 (Thu)	2	Completed	Conducted an extensive source code audit across all Module 1 packages, eliminating redundant variables, cleaning unused imports, and optimizing SQL strings. I authored comprehensive Javadoc comments on all public interfaces, service methods, and DAO implementations, explaining parameter contracts, return values, and thrown exceptions to facilitate seamless collaboration during the upcoming team integration phase.
04-09-2026 (Fri)	1.5	Completed	Committed the finalized Module 1 codebase to our shared GitHub repository following proper semantic commit messaging conventions. I updated our team project tracking spreadsheet, detailing completed servlet endpoints, DAO methods, and database schemas. I synchronized with peer module leads to confirm readiness for the full multi-module integration scheduled for the following week.

Week 10 - Integration of all five modules

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
07-09-2026 (Mon)	2	Completed	Initiated the multi-module project integration, assembling all five distinct student subsystems into a singular, unified Web Application Archive. I resolved package naming collisions, aligned shared model classes, and standardized Maven build configurations across the codebase. I verified that the combined project compiled cleanly without classpath conflicts or duplicate resource definitions in the build artifact.
08-09-2026 (Tue)	1.75	Completed	Connected the centralized authentication login controller to dynamically redirect authenticated users to their corresponding module landing pages. I verified that logging in as an Administrator loads the admin cockpit, Doctor credentials route to appointments, Receptionists reach patient registration, Nurses access vitals, and Patients open personal health portals accurately without routing errors.
09-09-2026 (Wed)	2	Completed	Conducted a thorough security filter review across all integrated servlet mappings and JSP resource paths. I audited URL pattern definitions in web.xml to confirm that newly merged endpoints from peer modules are strictly guarded by AuthenticationFilter and AuthorizationFilter, ensuring unauthenticated or unauthorized users cannot bypass access controls through forged direct URLs.
10-09-2026 (Thu)	2	Completed	Harmonized navigation bars, headers, and sidebars across all five modules to establish visual and functional consistency throughout the portal. I eliminated disparate CSS styles, standardized button designs, and corrected broken relative hyperlinks that emerged after merging distinct JSP directories. The global interface now presents a cohesive, professional hospital management user experience.
11-09-2026 (Fri)	1.5	Completed	Executed comprehensive end-to-end integration walkthroughs validating full user journey lifecycles across all five system roles. I verified that an administrator can provision doctor and receptionist accounts, receptionists can register patients, doctors can record diagnoses, and patients can view records. All transitions functioned reliably, confirming that core data flows across modules operate seamlessly.

Week 11 - Testing (unit, functional, role-based)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
14-09-2026 (Mon)	1.75	Completed	Designed and executed structured unit test suites targeting authentication workflows and edge cases. I created rigorous test scenarios evaluating correct credential logins, incorrect password rejections, non-existent username handling, and locked account responses. I confirmed that failed authentication attempts increment failed counters appropriately and return descriptive, secure feedback without revealing sensitive account existence details.
15-09-2026 (Tue)	2	Completed	Conducted extensive role-based security testing by manually attempting unauthorized URL access from different browser sessions. I verified that patient accounts attempting to access admin servlets were blocked with HTTP 403 Forbidden responses. Similarly, I confirmed that receptionists cannot access clinical consultation forms, validating that AuthorizationFilter enforces strict role segregation across all endpoints.
16-09-2026 (Wed)	1.75	Completed	Executed session lifecycle and security persistence tests under simulated idle and active usage scenarios. I validated that sessions expire precisely after thirty minutes of inactivity, terminating access and redirecting users to the login screen. Furthermore, I verified that manual logout invalidates session tokens immediately, preventing session reuse via browser history navigation or cached pages.
17-09-2026 (Thu)	2	Completed	Performed rigorous validation testing across administrative user registration and departmental management forms. I submitted boundary test inputs, including excessively long strings, special characters, malformed email addresses, and blank submissions. I verified that both client-side JavaScript and server-side ValidationUtil intercept invalid inputs, presenting clear, non-destructive error messages while preserving previously typed form data.
18-09-2026 (Fri)	1.75	Completed	Logged and documented all defects identified during testing in our centralized bug tracking log. I collaborated with team members to resolve high-priority issues, including fixing a session attribute collision and correcting an authorization bypass edge case on static assets. We re-tested all modified components to ensure complete regression stability across the platform.

Week 12 - Deployment (Docker, Render, cloud MySQL) and bug fixing

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
21-09-2026 (Mon)	1.75	Completed	Authored an enterprise-grade multi-stage Dockerfile to containerize the complete SmartCare hospital management application. I configured a Maven build container using OpenJDK 17 to compile application source code and generate the production WAR archive. For the runtime stage, I configured an official Apache Tomcat 10.1 image, optimizing container startup parameters and deploying the WAR artifact.
22-09-2026 (Tue)	1.75	Completed	Provisioned a managed cloud MySQL database instance on Aiven to host our production hospital data. I exported our normalized local database schema, migrated table definitions, and established relational integrity constraints in the cloud environment. I populated baseline administrative accounts, department listings, and essential lookup records, verifying reliable remote connectivity from our application environment.
23-09-2026 (Wed)	1.75	Completed	Updated db.properties configuration files to establish secure, encrypted JDBC connections with the remote cloud database. I enabled SSL connection parameters, configured connection pool bounds, and implemented environment variable overrides for database credentials. This ensures production credentials remain secure without being hardcoded into version control repositories, adhering strictly to modern security practices.
24-09-2026 (Thu)	2	Completed	Successfully deployed the containerized hospital web application to the Render cloud hosting platform. I connected our GitHub repository to Render, configured automatic web service builds, and mapped production environment variables. I verified that the deployed web service started cleanly, established database connections, and rendered the public login page flawlessly on the live public URL.
25-09-2026 (Fri)	1.5	Completed	Conducted comprehensive smoke testing on the live cloud deployment, verifying end-to-end functionality across desktop and mobile browsers. I identified and resolved a minor issue involving cross-origin cookie restrictions on cloud load balancers. I documented deployment procedures, production environment parameters, and container configurations in the deployment chapter of our team project report.

Week 13 - Report, PPT, demo video and viva preparation

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
28-09-2026 (Mon)	2	Completed	Authored the comprehensive introduction, project background, and technology stack sections of our final Project Based Learning report. I articulated the clinical and administrative motivations behind developing the Smart Hospital Management System. Furthermore, I detailed our selection of Java Servlets, JSP, MySQL, and Docker, justifying architectural decisions based on enterprise scalability and reliability benchmarks.
29-09-2026 (Tue)	2	Completed	Documented the comprehensive authentication, administrative control, and system security architectures within the technical report. I authored detailed explanations of BCrypt hashing algorithms, session hijacking prevention techniques, servlet filter chains, and RBAC authorization matrices. I inserted high-resolution architectural schematics and database entity diagrams to illustrate our implementation thoroughly for faculty examiners.
30-09-2026 (Wed)	2	Completed	Designed professional, high-impact presentation slides summarizing our hospital platform's architectural blueprint, security implementation, and administrative capabilities. I structured slide decks with concise technical bullet points, workflow diagrams, database relationship charts, and live system screenshots. I rehearsed presentation timing, ensuring our architectural highlights fit smoothly within the allotted faculty evaluation window.
01-10-2026 (Thu)	1.75	Completed	Recorded and edited a polished, high-definition video demonstration showcasing the hospital system's authentication and administrative features. I demonstrated administrator login, staff profile provisioning, departmental modifications, role permission enforcement, and real-time audit logging. I added concise audio commentary explaining underlying servlet and filter operations, producing a professional artifact for our portal submission.
02-10-2026 (Fri)	1.75	Completed	Conducted rigorous viva voce preparation sessions, reviewing theoretical principles and practical code implementations of servlet lifecycles, filter chaining, and cryptographic hashing. I rehearsed technical explanations regarding database connection pooling, session invalidation protocols, and Docker multi-stage builds. I finalized our submitted project package, ensuring complete readiness for our upcoming academic defense.

Total hours logged for Devansh Mathur: 115.75 hours over 65 days.

2. Deepak Kumar (24E1ARADM30P035)

Module 2 - Patient and Doctor Management

Member	Deepak Kumar (24E1ARADM30P035)
Module	Module 2 - Patient and Doctor Management
Code owned	Patient, Doctor, Receptionist models, PatientDAO, DoctorDAO, ReceptionistDAO, PatientServlet, DoctorServlet, patient JSPs (dashboard, profile, medical records), doctor JSPs (dashboard, patient profile, records), receptionist patient registration.
Period	06-07-2026 to 04-10-2026 (13 weeks). Daily hours between 1.5 and 2.0.
GitHub link	https://github.com/mathurdevansh/Smart-Hospital-Management-System

Part A - Weekly Reports (13 Weeks) — Deepak Kumar

Week 1 | 06-07-2026 to 12-07-2026 | Phase: Team formation, guide selection and abstract

Week dropdown	Week 1
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Collaborated with the project team to define the functional scope of the Patient and Doctor Management subsystem. Conducted field research analyzing operational workflows in local clinical environments, focusing on patient intake protocols and doctor record keeping. Formulated the module abstract detailing patient registration, personal health records, physician directories, and receptionist tools. Engaged in technical discussions with our faculty guide and team members to confirm the Java Servlet-JSP and MySQL architecture. Reviewed and polished the integrated project abstract, successfully submitting it to the portal.
Problems Faced	Team had many different ideas. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Listed patient requirements: registration, profile, history and records. (2) Listed doctor requirements: profile, department, specialisation and availability. (3) Wrote data requirements for patient and doctor details.

Week 2 | 13-07-2026 to 19-07-2026 | Phase: SRS (Software Requirement Specification)

Week dropdown	Week 2
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Formulated detailed functional and non-functional software requirement specifications for the Patient and Doctor Management module. Cataloged comprehensive requirements for patient onboarding, dynamic profile updating, medical history archival, and doctor departmental scheduling. Established critical data integrity constraints including system-generated unique patient identifiers and mandatory medical field validations. Specified non-functional benchmarks covering responsive interface loading, database query throughput, and data privacy protection. Consolidated the documentation into the team Software Requirement Specification and conducted a comprehensive review with peers to guarantee functional completeness.
Problems Faced	Overlap between patient and receptionist features. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Drew the use-case diagram for Patient and Doctor actors. (2) Designed the class diagram for Patient with attributes and methods. (3) Designed the class diagram for Doctor and Receptionist.

Week 3 | 20-07-2026 to 26-07-2026 | Phase: UML design (use-case, class, sequence)

Week dropdown	Week 3
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Constructed detailed Unified Modeling Language diagrams mapping the structural and behavioral architecture of the patient and doctor subsystem. Formulated use-case diagrams defining interaction boundaries for Patient, Doctor, and Receptionist actors within the system. Designed comprehensive class diagrams for Patient, Doctor, and Receptionist domain models with explicit attribute encodings, methods, and relationships. Engineered dynamic sequence diagrams detailing the multi-tier registration sequence from receptionist form submission to database persistence. Merged all module diagrams into the master project design document following mentor guidelines.
Problems Faced	Deciding attributes for Patient class. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Designed the patients table with personal and contact fields. (2) Designed the doctors table with department link and qualification. (3) Designed the medical_records table and its relation to patients.

Week 4 | 27-07-2026 to 02-08-2026 | Phase: Database (ER) design, UI mock-ups, project setup

Week dropdown	Week 4
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Architected normalized relational database schemas and initialized the local development environment for Module 2. Designed the patients relational table capturing comprehensive demographic, emergency contact, and medical baseline fields with primary key constraints. Structured the doctors table with foreign key references to hospital departments, specializations, and qualifications. Modeled the medical_records entity linked directly to patient identifiers. Installed local JDK 17, Tomcat, and MySQL environments, cloned the shared Git repository, and drafted user interface mock-ups for patient registration and clinical dashboards.
Problems Faced	Normalising patient and record tables. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote Patient model class with all fields. (2) Wrote Doctor and Receptionist model classes. (3) Implemented PatientDAO insert and select methods.

Week 5 | 03-08-2026 to 09-08-2026 | Phase: Module coding - models and DAO layer

Week dropdown	Week 5
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Engineered core domain model classes and foundational Data Access Object implementations for patient and doctor operations. Authored Patient, Doctor, and Receptionist model beans adhering to strict encapsulation standards with complete accessors and mutators. Implemented PatientDAO containing JDBC PreparedStatement routines for inserting patient profiles, retrieving records by identifier, and executing multi-criteria searches. Developed DoctorDAO providing departmental filtering and physician availability lookups. Verified all transactional queries against the local MySQL instance to ensure consistent data persistence and eliminate SQL syntax errors.
Problems Faced	Date conversion between java.util and java.sql. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Implemented PatientServlet actions for view and update profile. (2) Implemented DoctorServlet dashboard action with today's data. (3) Added the patient search action by name and phone.

Week 6 | 10-08-2026 to 16-08-2026 | Phase: Module coding - services and servlets

Week dropdown	Week 6
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Developed business service logic and HTTP servlet controllers managing patient profile modifications and clinical dashboards. Built PatientServlet handling profile viewing, demographic updates, and medical history retrieval routines. Implemented DoctorServlet providing personalized daily dashboard summaries of assigned consultations and departmental schedules. Created patient search handlers facilitating rapid lookups by name and contact phone number. Built ReceptionistDAO and integrated patient registration controller actions, executing rigorous test runs with mock clinical data to confirm error-free servlet request dispatching and session management.
Problems Faced	Null pointer when patient had no record. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Created the patient dashboard.jsp with quick links. (2) Created the patient profile.jsp with editable form. (3) Created the doctor dashboard.jsp with appointment summary.

Week 7 | 17-08-2026 to 23-08-2026 | Phase: Module coding - JSP views and UI

Week dropdown	Week 7
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Constructed responsive JavaServer Pages delivering interactive user interfaces for patients, doctors, and hospital receptionists. Developed patient_dashboard.jsp featuring intuitive navigation tiles, upcoming appointments, and quick medical history links. Built patient_profile.jsp with client-side editable forms and validation feedback banners. Developed doctor_dashboard.jsp presenting clinical queues alongside doctor patient_profile.jsp displaying comprehensive historical diagnoses. Engineered register_patient.jsp and patient listing interfaces for hospital receptionists, integrating reusable navigation headers, dynamic sidebars, and external style sheets to ensure consistent visual aesthetics and user experience across all module views.
Problems Faced	Form values lost after validation failure. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Added server-side validation for phone, age and blood group. (2) Added JavaScript validation on registration form fields. (3) Handled exceptions in DAO methods with clear user messages.

Week 8 | 24-08-2026 to 30-08-2026 | Phase: Module coding - validation, exceptions, security

Week dropdown	Week 8
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Implemented rigorous server-side input validation, client-side scripting controls, and exception handling protocols across all module interfaces. Added server-side validation routines in servlets verifying telephone numbers, age boundaries, and valid blood group classifications. Developed client-side JavaScript checks preventing malformed form submissions before reaching backend controllers. Refactored DAO methods with descriptive custom exceptions and user-friendly error banners. Incorporated JSTL tag libraries with proper cross-site scripting escaping, and implemented pagination mechanisms across large patient records tables to maintain optimal interface responsiveness.
Problems Faced	Search became slow without proper query. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Completed the medical records view for the patient. (2) Added the doctor's records.jsp to view and add notes. (3) Added patient auto-ID generation.

Week 9 | 31-08-2026 to 06-09-2026 | Phase: Module coding - module completion and self-review

Week dropdown	Week 9
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Finalized medical record visualization components and completed rigorous module refinement and code polishing prior to integration. Built the comprehensive medical records view enabling patients to inspect longitudinal clinical summaries and diagnostic histories. Developed doctor's records.jsp interface allowing attending physicians to review past treatments and append consultation progress notes. Implemented automated unique patient identification generation routines ensuring collision-free numbering. Conducted thorough code reviews, refactored data access methods, inserted comprehensive Javadoc documentation, and pushed clean module code to the team repository.
Problems Faced	Time was short for comments and cleanup. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Integrated patient registration with the login of the patient user. (2) Linked doctor listing with the appointment booking page. (3) Checked navigation between patient and doctor pages after merge.

Week 10 | 07-09-2026 to 13-09-2026 | Phase: Integration of all five modules

Week dropdown	Week 10
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Participated actively in the multi-module system integration, unifying patient and doctor management with peer hospital subsystems. Integrated receptionist patient onboarding workflows with automated user account creation and credential assignment in Module 1. Linked physician directories with appointment booking and scheduling interfaces in Module 3. Validated smooth navigation flows between patient directories and clinical consultation forms. Resolved database connection leaks identified during continuous integration testing, and executed comprehensive end-to-end user journeys from initial patient registration to complete profile inspection.
Problems Faced	Connections were not closed in a few methods. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote test cases for patient registration and profile update. (2) Tested invalid and boundary inputs on forms. (3) Tested doctor dashboard data with multiple doctors.

Week 11 | 14-09-2026 to 20-09-2026 | Phase: Testing (unit, functional, role-based)

Week dropdown	Week 11
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Executed thorough unit, functional, and boundary testing across all patient and doctor management components. Developed structured test cases covering patient registration, demographic updates, and doctor profile management under diverse clinical scenarios. Tested input boundary limits on web forms, verifying robust rejection of negative ages, malformed phone numbers, and invalid characters. Evaluated doctor dashboard metrics across multiple concurrent physician sessions. Conducted search and pagination stress tests on large patient datasets, logging identified defects and verifying fixes through exhaustive regression retesting.
Problems Faced	Some inputs allowed negative age. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Checked patient and doctor pages on the deployed site. (2) Fixed date format issue on the cloud database. (3) Added missing seed data for departments and doctors.

Week 12 | 21-09-2026 to 27-09-2026 | Phase: Deployment (Docker, Render, cloud MySQL) and bug fixing

Week dropdown	Week 12
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Coordinated cloud deployment testing and resolved platform-specific database discrepancies on the live production server. Verified all patient and doctor management JSP interfaces and servlet controllers within the Dockerized container running on Render. Identified and corrected date formatting inconsistencies arising between local MySQL and cloud-hosted database engines. Injected essential seed data populating realistic medical departments and doctor profiles for demonstration purposes. Retested registration workflows on the live server and resolved responsive layout anomalies on compact mobile screens to guarantee complete cross-device usability.
Problems Faced	Date format differed on cloud MySQL. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote the patient and doctor module section of the report. (2) Added screenshots and table descriptions to the report. (3) Prepared module slides for the presentation.

Week 13 | 28-09-2026 to 04-10-2026 | Phase: Report, PPT, demo video and viva preparation

Week dropdown	Week 13
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Compiled technical documentation, recorded module demonstrations, and completed comprehensive viva preparation for academic evaluation. Authored the dedicated patient and doctor management chapter of the final project report, detailing entity designs, DAO implementations, and architectural workflows. Formulated high-resolution schema diagrams, UI screenshots, and relational table specifications. Designed presentation slides highlighting clinical data modeling and role-based workflows. Rehearsed live system demonstrations of patient registration, and prepared detailed technical answers on JDBC transaction handling connection pooling, and DAO design patterns for our upcoming viva defense.
Problems Faced	Report formatting took extra time. Solved with team discussion and guide's suggestions.
Plan for Next Week	Final viva and demo, submit report, PPT, SRS, demo video and working software on the portal.

Part B - Daily Work Log (65 Entries) — Deepak Kumar

Status for all entries: Completed. Saturday is the weekly mentor meeting, so no daily entry is listed.

Week 1 - Team formation, guide selection and abstract

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
06-07-2026 (Mon)	2	Completed	Joined all five team members during our project kickoff discussion to identify critical problems in healthcare operations that software could solve. We evaluated manual registration bottlenecks, misplaced patient files, and uncoordinated doctor schedules across outpatient clinics. We reached consensus on building the Smart Hospital Management System, demarcated module responsibilities, and initiated collaborative project planning.
07-07-2026 (Tue)	1.75	Completed	Conducted targeted field research to investigate how patient intake, demographic recording, and physician assignment workflows operate within regional medical clinics. I documented traditional registration desk practices, physical filing systems, and doctor scheduling registries, identifying frequent administrative errors and manual tracking inefficiencies. This operational analysis established foundational data requirements for automating patient and doctor management.
08-07-2026 (Wed)	1.5	Completed	Authored the specialized abstract section detailing patient registration workflows, longitudinal medical history management, and doctor directory services. I outlined the core architectural objectives of Module 2, emphasizing how digitized records eliminate patient record duplication, streamline clinical handoffs, and enhance patient data privacy. I aligned the technical narrative with the overarching SmartCare project vision.
09-07-2026 (Thu)	2	Completed	Participated in a technical consultation with our faculty project guide, Er. Ram Babu Buri, and teammates to evaluate enterprise technology alternatives. We confirmed using Java Servlets, JSP, JSTL, and MySQL as our core development stack. We discussed database normalization standards, session tracking mechanisms, and development milestone schedules to ensure structured delivery.
10-07-2026 (Fri)	1.75	Completed	Reviewed and refined the consolidated project abstract in collaboration with all module leads, verifying that patient onboarding and doctor management interfaces integrated cleanly with administrative and billing specifications. We incorporated our guide's structural feedback, completed final document proofreading, and submitted the finalized abstract proposal through the university project portal successfully.

Week 2 - SRS (Software Requirement Specification)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
13-07-2026 (Mon)	1.5	Completed	Documented exhaustive functional requirements for the patient subsystem within our formal Software Requirement Specification. I cataloged explicit operational criteria governing patient onboarding, personal demographic profile modifications, emergency contact records, and longitudinal medical history retrieval. I defined input expectations for blood groups, existing allergies, and chronic condition notes to support clinical safety.
14-07-2026 (Tue)	1.75	Completed	Specified comprehensive functional requirements for doctor management and clinical scheduling within the SRS document. I detailed specifications for managing doctor professional profiles, departmental affiliations, medical specializations, consultation chamber assignments, and weekly visiting schedules. These requirements ensure receptionist desks and appointment modules maintain access to accurate, up-to-date physician availability data across the facility.
15-07-2026 (Wed)	2	Completed	Formulated detailed data entity requirements for patient and doctor records, establishing exact data types, precision limits, and database constraints. I specified fields for patient names, contact numbers, residential addresses, emergency contacts, doctor license numbers, and consultation fees. This documentation provided unambiguous guidelines for upcoming relational database modeling and input validation logic.
16-07-2026 (Thu)	2	Completed	Established critical system constraints and validation rules within the specification document to preserve medical record integrity. I mandated the automated generation of unique alphanumeric patient identifiers to eliminate duplicate profiles. Additionally, I designated mandatory demographic fields, phone number format constraints, and age validity boundaries to prevent malformed patient entries across forms.
17-07-2026 (Fri)	1.75	Completed	Consolidated the patient and doctor management specifications into the team's master Software Requirement Specification document. I led an internal review walkthrough with team members, clarifying functional handoffs between receptionist registration actions and patient self-service profiles. We harmonized terminologies, resolved minor workflow overlaps, and finalized the formal SRS baseline document with complete team approval.

Week 3 - UML design (use-case, class, sequence)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
20-07-2026 (Mon)	1.75	Completed	Engineered comprehensive Unified Modeling Language use-case diagrams for Patient and Doctor actors using Visual Paradigm. I mapped functional use cases including registering new patients, updating medical profiles, reviewing diagnostic records, managing doctor availability, and viewing assigned patient queues. I accurately established include and extend associations representing prerequisite authentication checks and validation dependencies.
21-07-2026 (Tue)	2	Completed	Designed the structural class diagram for the Patient domain model, establishing encapsulated attributes, constructor signatures, and business methods. I modeled fields including patientId, personal details, contact information, bloodGroup, and emergencyContacts, accompanied by standard getters and setters. I outlined relational associations connecting patient profiles to medical records and scheduled appointment entities.
22-07-2026 (Wed)	2	Completed	Constructed detailed class diagrams for the Doctor and Receptionist entities, defining their internal structures and associations with hospital departments. For the Doctor model, I included specialization, departmentId, and qualification attributes, while the Receptionist class encapsulated desk assignment identifiers. I established foreign key associations linking doctors to medical departments and consultation records.
23-07-2026 (Thu)	1.75	Completed	Drafted an intricate sequence diagram modeling the dynamic runtime workflow of registering a new patient into the hospital system. The diagram captures message exchanges between the receptionist browser client, the front controller servlet, server-side validation utilities, the patient DAO layer, and the MySQL database. I documented both successful persistence and error rollback scenarios.
24-07-2026 (Fri)	1.5	Completed	Merged all Module 2 UML diagrams into the master team design specification document. I collaborated with the Module 1 and Module 3 leads to reconcile class attribute namings, ensuring seamless cross-module referencing between User credentials, Doctor profiles, and Appointment bookings. We validated the consolidated design against our mentor's modeling guidelines.

Week 4 - Database (ER) design, UI mock-ups, project setup

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
27-07-2026 (Mon)	1.75	Completed	Designed the relational database schema for the patients table in MySQL, applying normalization principles up to Third Normal Form. I defined columns for patient identification, full name, date of birth, gender, blood group, contact phone, and residential address. I established primary key constraints, unique phone indexes, and audit timestamps to ensure data reliability.
28-07-2026 (Tue)	1.5	Completed	Architected the relational schema for the doctors table and configured foreign key linkages to hospital departments. I incorporated fields for medical registration numbers, educational qualifications, clinical specializations, consultation fees, and visiting hours. I established cascading constraints and indexed department foreign keys to optimize query performance during receptionist physician lookup operations.
29-07-2026 (Wed)	1.75	Completed	Designed the medical_records relational table schema to establish a longitudinal clinical history repository for registered patients. I structured columns for record identification, patient foreign key, diagnosing physician reference, recorded symptoms, clinical diagnosis, and visit date. I established strict foreign key constraints preventing orphan records if parent patient profiles are modified.
30-07-2026 (Thu)	2	Completed	Configured my local software engineering environment by installing Java Development Kit 17, Apache Tomcat 10.1 web server, and MySQL Community Server 8.0. I cloned the centralized team GitHub repository, imported the Maven project structure into my IDE, and successfully compiled and deployed the baseline sample web application on the local server.
31-07-2026 (Fri)	1.5	Completed	Created intuitive, high-fidelity wireframes and user interface mock-ups for patient registration forms and doctor dashboard views. Using modern layout tools, I designed responsive form layouts incorporating organized fieldsets for personal, medical, and emergency contact details. For the doctor dashboard, I structured clean visual tables displaying scheduled appointments and patient summaries.

Week 5 - Module coding - models and DAO layer

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
03-08-2026 (Mon)	1.5	Completed	Authored the complete Patient JavaBean domain model class in Java, strictly implementing standard encapsulation principles. I declared private instance variables for all demographic, medical, and contact attributes, providing comprehensive parameterized and default constructors. I implemented standard getter and setter methods, alongside overridden toString, equals, and hashCode methods for robust collection management.
04-08-2026 (Tue)	1.5	Completed	Implemented the Doctor and Receptionist JavaBean model classes in Java to represent hospital medical and administrative staff. I defined encapsulated fields for professional qualifications, departmental mappings, licensing details, and desk assignments. I wrote clean accessor and mutator methods, ensuring seamless data transfer between HTTP servlet request parameters and backend data persistence layers.
05-08-2026 (Wed)	1.5	Completed	Constructed the PatientDAO interface and developed fundamental database persistence methods using JDBC. I implemented insertPatient and getPatientById routines utilizing PreparedStatement to guarantee protection against SQL injection vulnerabilities. I mapped database query ResultSet rows directly into instantiated Patient model objects, managing connection lifetimes safely using Java try-with-resources structural blocks to ensure stability.
06-08-2026 (Thu)	2	Completed	Expanded PatientDAO by programming advanced data manipulation routines, including updatePatientProfile, searchPatientsByNameOrPhone, and getAllPatientsWithPagination. I authored parameterized SQL queries that support dynamic filtering while maintaining optimal query execution speeds. I tested these methods against our local MySQL test database, verifying accurate data modifications and robust exception handling across all test cases.
07-08-2026 (Fri)	1.5	Completed	Implemented the DoctorDAO class providing comprehensive query methods to retrieve physician profiles based on departmental classifications and availability status. I authored parameterized SQL statements to fetch all active doctors, retrieve physician profiles by department ID, and update contact details. I verified query correctness using standalone test scripts on the local database.

Week 6 - Module coding - services and servlets

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
10-08-2026 (Mon)	1.5	Completed	Programmed the PatientServlet controller to manage incoming HTTP GET and POST requests regarding patient profile operations. In the doGet handler, the servlet retrieves patient records and forwards them to profile view pages. In the doPost handler, it extracts updated demographic parameters, validates inputs, invokes PatientDAO, and redirects users with status messages.
11-08-2026 (Tue)	1.75	Completed	Developed the DoctorServlet controller to power personalized clinical dashboards for attending physicians. I implemented request handling logic that extracts the logged-in doctor's unique identifier from the active HTTP session, queries AppointmentDAO and PatientDAO, and aggregates the day's scheduled consultations, waiting room queues, and recent clinical notes onto the dashboard model.
12-08-2026 (Wed)	1.75	Completed	Engineered dedicated patient search controller actions within PatientServlet to support rapid front-desk patient lookups. I implemented query handling that processes search strings containing patient names, registration identifiers, or phone numbers. The servlet delegates lookup requests to PatientDAO and returns matching result sets formatted for display on receptionist intake screens smoothly.
13-08-2026 (Thu)	2	Completed	Constructed the ReceptionistDAO class and integrated receptionist patient intake actions into the application controllers. I programmed database operations to record staff registration activities and link registered patients to system user accounts. I verified that newly registered patient records trigger automated account provisioning, ensuring immediate portal access for onboarded individuals without delays.
14-08-2026 (Fri)	1.5	Completed	Executed comprehensive unit and integration testing across PatientServlet and DoctorServlet controller actions using realistic mock clinical datasets. I tested varied input parameters, verified correct session context extraction, and inspected HTTP redirection headers. I validated that all servlet endpoints handle missing or invalid query parameters gracefully without throwing unhandled container exceptions.

Week 7 - Module coding - JSP views and UI

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
17-08-2026 (Mon)	2	Completed	Developed the user-facing patient_dashboard.jsp view using semantic HTML5 and modern responsive CSS flexbox layouts. I structured an intuitive patient portal featuring quick-action shortcut tiles, upcoming appointment summary cards, active prescription notifications, and direct navigation links to personal health records. I ensured the interface scales cleanly across mobile, tablet, and desktop viewports.
18-08-2026 (Tue)	1.75	Completed	Built the interactive patient_profile.jsp view allowing registered patients to inspect and update their personal contact information. I structured intuitive form controls for residential address, contact numbers, and emergency contacts, keeping core medical identifiers read-only. I integrated dynamic JSTL alert boxes to render success confirmation banners or server-side validation error messages clearly.
19-08-2026 (Wed)	1.75	Completed	Constructed the doctor_dashboard.jsp view delivering an efficient clinical workspace for hospital physicians. I designed a structured overview panel summarizing daily patient appointments, scheduled consultation time slots, pending lab results, and inpatient admission counts. I incorporated real-time status badges, enabling doctors to monitor patient arrival statuses at a single glance with clarity.
20-08-2026 (Thu)	2	Completed	Developed the doctor patient_profile.jsp interface providing attending physicians with immediate access to comprehensive clinical histories. I designed clear tabular sections presenting chronological diagnostic notes, prior medication prescriptions, recorded vital signs, and past hospital visits. This unified historical view empowers doctors to make well-informed clinical diagnoses during patient consultations across hospital wards.
21-08-2026 (Fri)	2	Completed	Engineered the receptionist register_patient.jsp and patient listing views to optimize front-desk intake operations. I designed a structured registration form grouping demographic, contact, and medical baseline fields cleanly. Furthermore, I built the patient listing table with search filters and quick-action buttons, integrating common navigation headers and sidebars for visual consistency throughout.

Week 8 - Module coding - validation, exceptions, security

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
24-08-2026 (Mon)	1.5	Completed	Implemented rigorous server-side validation rules within PatientServlet and reception controllers to ensure clinical data integrity. I authored validation checks verifying ten-digit telephone formats, positive age values, valid calendar birthdates, and standardized ABO blood group notations. Form submissions containing malformed values are halted immediately and returned with field-specific guidance messages to users.
25-08-2026 (Tue)	1.5	Completed	Developed robust client-side JavaScript validation scripts to provide instant feedback on registration and profile editing forms. I added event listeners that check input formatting as users type, highlighting invalid fields with clear visual cues before form submission. This dual-layer validation improves user experience while preventing malformed payloads from consuming server resources.
26-08-2026 (Wed)	2	Completed	Refactored exception handling across all PatientDAO and DoctorDAO methods to guarantee graceful error management. I wrapped low-level SQLExceptions into custom DatabaseException instances containing user-friendly diagnostic messages while logging technical stack traces to server logs. This ensures web interfaces display intelligible error notices without exposing internal database structures to users during operations.
27-08-2026 (Thu)	2	Completed	Enhanced security across all Module 2 JavaServer Pages by implementing standard JSTL core tag libraries with strict output escaping. I refactored raw expression outputs into c:out tags, effectively mitigating Cross-Site Scripting vulnerabilities by sanitizing user-supplied strings. I verified that medical notes containing special characters render cleanly without executing injected scripts.
28-08-2026 (Fri)	1.5	Completed	Implemented server-side pagination mechanisms across the central patient directory listing interface. I modified SQL queries in PatientDAO to utilize LIMIT and OFFSET clauses, calculating total page counts dynamically. I added intuitive pagination navigation controls on the JSP view, ensuring smooth page transitions and rapid response times even with extensive patient populations.

Week 9 - Module coding - module completion and self-review

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
31-08-2026 (Mon)	2	Completed	Finalized the longitudinal medical records visualization interface for registered patients in patient_records.jsp. I structured a chronological clinical timeline displaying past diagnostic summaries, attending physician notes, and consultation dates. I formatted medical timestamps cleanly using JSTL format tags, allowing patients to review their comprehensive treatment histories conveniently from home with total ease.
01-09-2026 (Tue)	2	Completed	Developed the doctor's records.jsp interface enabling physicians to inspect prior consultations and record fresh clinical progress notes during active examinations. I created structured text areas for chief complaints, clinical observations, provisional diagnoses, and recommended therapies. The interface persists consultation notes directly into the medical_records table via dedicated servlet actions reliably.
02-09-2026 (Wed)	1.5	Completed	Engineered an automated unique patient identification numbering algorithm within PatientDAO. The generator constructs standardized alphanumeric identifiers incorporating registration years and incremental numeric sequences. I implemented concurrency-safe database counter lookups to prevent duplicate identifier collisions during simultaneous front-desk patient registrations, ensuring robust uniqueness across the hospital system during high-traffic clinical operations.
03-09-2026 (Thu)	1.5	Completed	Conducted a comprehensive code cleanup and peer review across all Module 2 Java packages. I refactored inconsistent method names, removed deprecated database queries, and eliminated unused imports. Additionally, I authored detailed Javadoc comments on all model classes, DAO interfaces, and servlet handlers, documenting parameter requirements and return types to prepare for integration.
04-09-2026 (Fri)	1.75	Completed	Committed the finalized, fully commented Module 2 source code to the shared team GitHub repository following standard Git branch conventions. I updated our shared progress tracking sheet with completed feature checklists, verified test coverage across DAO methods, and coordinated with peer module leads to confirm readiness for next week's system assembly.

Week 10 - Integration of all five modules

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
07-09-2026 (Mon)	2	Completed	Integrated the receptionist patient registration workflow with the central authentication system developed in Module 1. I configured registration handlers so that saving a new patient profile automatically provisions a corresponding User login record with encrypted default credentials. I validated that freshly onboarded patients can immediately authenticate and access their personal portal accounts.
08-09-2026 (Tue)	1.75	Completed	Linked the active physician directory and departmental listings with the appointment scheduling subsystem built in Module 3. I ensured that appointment booking forms query DoctorDAO dynamically to display available doctors categorized by medical specialization. This seamless integration enables patients and receptionists to select certified physicians for consultation bookings without synchronization delays.
09-09-2026 (Wed)	1.5	Completed	Audited navigation flows and UI transitions between patient management screens, doctor dashboards, and appointment schedules following the initial module merge. I corrected broken relative path links, aligned shared CSS class names, and verified that breadcrumb navigation links accurately reflect the user's role and hierarchical position across all application views cleanly.
10-09-2026 (Thu)	1.5	Completed	Investigated and resolved database connection leaks identified during continuous integration testing under concurrent session workloads. I audited every JDBC block in PatientDAO and DoctorDAO, ensuring all Connection, PreparedStatement, and ResultSet objects are strictly enclosed within try-with-resources statements to guarantee immediate resource release after query execution throughout the entire application lifecycle.
11-09-2026 (Fri)	2	Completed	Executed comprehensive end-to-end integration walkthroughs validating the entire patient lifecycle from front-desk intake to clinical review. I verified that a receptionist can register a new patient, the patient can log in and view their profile, and an assigned doctor can inspect the patient's record, confirming flawless data flow across integrated modules.

Week 11 - Testing (unit, functional, role-based)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
14-09-2026 (Mon)	2	Completed	Designed and executed structured unit and functional test cases evaluating patient registration and profile updating routines. I formulated test suites covering standard registrations, profile updates, and doctor queue retrievals, validating that all database transactions execute reliably and persist updated demographic data accurately without data truncation or character encoding errors on forms.
15-09-2026 (Tue)	1.5	Completed	Performed extensive boundary and negative input testing across patient intake forms. I submitted boundary test cases, including negative ages, non-numeric telephone strings, blank mandatory fields, and excessively long address descriptions. I verified that validation handlers successfully intercept invalid entries, displaying helpful warning banners without crashing or corrupting database records at all.
16-09-2026 (Wed)	2	Completed	Conducted rigorous functional testing on doctor dashboard views across multiple simulated physician profiles. I populated test schedules with varying appointment densities, verifying that each doctor's dashboard displays only their assigned patient queues and consultation records. I confirmed that departmental statistics and daily counts calculate accurately across concurrent doctor logins under load.
17-09-2026 (Thu)	1.75	Completed	Executed search algorithm and pagination stress tests on our patient directory using a simulated test database containing hundreds of patient records. I evaluated search responsiveness when filtering by partial names and telephone fragments, confirming that indexed database queries return accurate matching records within acceptable sub-second response thresholds consistently across queries.
18-09-2026 (Fri)	2	Completed	Documented all software defects uncovered during testing in our centralized issue tracker and collaborated with teammates to implement immediate code fixes. I resolved a minor date parsing defect on patient birthdates, retested all affected components, and verified complete regression stability across the patient and doctor management codebase with positive results.

Week 12 - Deployment (Docker, Render, cloud MySQL) and bug fixing

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
21-09-2026 (Mon)	1.5	Completed	Performed live deployment verification of patient and doctor management components on the Render cloud hosting environment. I accessed the live public application URL, navigated through patient registration and physician directory views, and verified that containerized Tomcat instances serve all JSP assets, images, and external stylesheets correctly without broken links on production.
22-09-2026 (Tue)	1.75	Completed	Investigated and resolved a date formatting defect observed on the cloud MySQL database server. The remote database engine defaulted to strict UTC timestamps, causing discrepancies with localized birthdates. I standardized date formatting routines within PatientDAO using ISO-8601 string conversions, ensuring consistent date rendering across local development and production cloud tiers.
23-09-2026 (Wed)	1.5	Completed	Authored and executed comprehensive SQL seed scripts on the cloud database instance to populate realistic demonstration data. I inserted sample medical departments, certified doctor profiles with varied specializations, and baseline patient records. This verified that production database constraints, foreign key cascades, and initial lookup tables operate reliably in the cloud.
24-09-2026 (Thu)	1.5	Completed	Conducted comprehensive retesting of the patient registration workflow directly on the live cloud production environment. I registered multiple test patient profiles through the receptionist interface, verified automated database persistence in cloud MySQL, and confirmed that confirmation flash messages and redirected views render smoothly on live web browsers across testing sessions.
25-09-2026 (Fri)	1.5	Completed	Audited and optimized user interface responsiveness across patient and doctor screens on various physical devices, including smartphones and tablets. I refined CSS media queries, adjusted table overflow behaviors, and resolved minor text wrapping issues on small viewports. I documented all deployment configurations and troubleshooting steps in the project report comprehensively.

Week 13 - Report, PPT, demo video and viva preparation

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
28-09-2026 (Mon)	1.5	Completed	Authored the complete Patient and Doctor Management chapter of our team's final Project Based Learning technical report. I documented module architecture, domain entity relationships, Data Access Object implementations, and servlet request dispatching lifecycles. I explained how our design ensures data normalization, patient confidentiality, and efficient clinical record retrieval throughout clinical workflows.
29-09-2026 (Tue)	2	Completed	Captured high-resolution user interface screenshots and authored descriptive table walkthroughs for inclusion in the project report. I illustrated patient registration forms, interactive medical dashboards, doctor queue tables, and clinical history timelines. I annotated key UI elements and aligned figure numbering with report text to present a clear, scholarly deliverable to evaluators.
30-09-2026 (Wed)	2	Completed	Designed professional, high-impact presentation slides summarizing Module 2 capabilities for our upcoming academic project defense. I structured slides covering clinical problem statements, entity relationship models, servlet architecture, and security validation layers. I incorporated clear visual diagrams and live system screenshots to deliver an engaging, informative presentation to faculty members during the final evaluation.
01-10-2026 (Thu)	1.75	Completed	Rehearsed live system demonstration workflows showcasing patient registration, doctor dashboard updates, and medical record viewing. I practiced smooth handoffs between receptionist intake actions and doctor diagnostic note entries, timing the presentation to fit within the allotted evaluation window. I verified that all demonstration accounts functioned reliably on the cloud server.
02-10-2026 (Fri)	1.5	Completed	Completed rigorous viva voce preparation focusing on core Java enterprise concepts, JDBC transaction management, and Data Access Object patterns. I rehearsed technical explanations regarding PreparedStatement advantages over raw SQL, MVC architectural separation, and session handling mechanics. I organized project source files, ensuring complete readiness for final evaluation during the viva defense.

Total hours logged for Deepak Kumar: 113.25 hours over 65 days.

3. Hansraj Pandey (24E1ARADM40P055)

Module 3 - Appointment and Consultation

Member	Hansraj Pandey (24E1ARADM40P055)
Module	Module 3 - Appointment and Consultation
Code owned	Appointment model, AppointmentDAO, AppointmentService (double-booking check), AppointmentServlet, MedicalRecord model and DAO, patient book_appointment/appointments JSPs, receptionist book_appointment/appointments JSPs, doctor appointments and consultation JSPs.
Period	06-07-2026 to 04-10-2026 (13 weeks). Daily hours between 1.5 and 2.0.
GitHub link	https://github.com/mathurdevansh/Smart-Hospital-Management-System

Part A - Weekly Reports (13 Weeks) — Hansraj Pandey

Week 1 | 06-07-2026 to 12-07-2026 | Phase: Team formation, guide selection and abstract

Week dropdown	Week 1
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Collaborated with team members during the project kickoff to delineate functional boundaries for the Appointment and Consultation subsystem. Researched outpatient clinic scheduling protocols, focusing on queue bottlenecks, double-booking occurrences, and manual consultation logging issues. Authored the dedicated abstract section detailing automated slot reservation, doctor appointment queues, and electronic medical diagnostic records. Consulted with our faculty mentor and teammates to validate the Java Servlet, JSP, and MySQL architectural foundation. Completed thorough review sessions and submitted the unified abstract proposal to the portal.
Problems Faced	Choosing the right scope for the idea. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Listed functional requirements for booking, rescheduling and cancelling. (2) Wrote requirements for doctor consultation and diagnosis notes. (3) Defined appointment statuses: Scheduled, Completed and Cancelled.

Week 2 | 13-07-2026 to 19-07-2026 | Phase: SRS (Software Requirement Specification)

Week dropdown	Week 2
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Authored comprehensive functional and non-functional requirement specifications for the Appointment and Consultation module within the team SRS document. Outlined detailed procedural criteria governing appointment booking, date-time rescheduling, and patient cancellations. Defined explicit consultation workflows allowing physicians to document clinical symptoms, provisional diagnoses, and follow-up directives. Codified critical business logic rules strictly prohibiting concurrent double-booking of physician consultation time slots. Consolidated all appointment specifications into the team SRS document and conducted an architectural walkthrough with teammates to verify functional completeness.
Problems Faced	Defining all appointment statuses clearly. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Drew the use-case diagram for booking and consultation. (2) Designed the class diagram for Appointment. (3) Designed the class diagram for MedicalRecord.

Week 3 | 20-07-2026 to 26-07-2026 | Phase: UML design (use-case, class, sequence)

Week dropdown	Week 3
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Engineered comprehensive Unified Modeling Language diagrams mapping the structural and behavioral architecture of the appointment and consultation subsystem. Developed use-case diagrams defining interaction boundaries for Patient, Receptionist, and Doctor actors. Designed structural class diagrams for Appointment and MedicalRecord domain models detailing encapsulated attributes, constructor definitions, and operational methods. Constructed dynamic sequence diagrams capturing the complete booking verification sequence, specifically illustrating conflict checking logic within slot repositories. Merged all module diagrams into the master project design document following mentor guidelines.
Problems Faced	Showing the double-booking rule in sequence diagram. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Designed the appointments table with doctor, patient and slot columns. (2) Designed the medical_records table linked to appointments. (3) Worked out indexes for date and doctor lookups.

Week 4 | 27-07-2026 to 02-08-2026 | Phase: Database (ER) design, UI mock-ups, project setup

Week dropdown	Week 4
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Architected normalized relational database schemas and set up the local development environment for Module 3. Designed the appointments relational table incorporating doctor references, patient foreign keys, date-time slots, and status fields with primary key constraints. Structured the medical_records table establishing direct relational linkages to appointment identifiers. Engineered composite indexes across appointment date and doctor columns to optimize lookup performance. Installed local JDK 17, Tomcat, and MySQL environments, cloned the shared Git repository, and drafted user interface mock-ups for booking forms.
Problems Faced	Choosing correct indexes for lookups. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote the Appointment model with date and time fields. (2) Wrote the MedicalRecord model class. (3) Implemented AppointmentDAO insert and fetch by patient.

Week 5 | 03-08-2026 to 09-08-2026 | Phase: Module coding - models and DAO layer

Week dropdown	Week 5
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Implemented foundational data access layer components and domain models for appointment scheduling and clinical consultation records. Authored Appointment and MedicalRecord Java model classes strictly adhering to JavaBean encapsulation standards with complete accessors and mutators. Developed AppointmentDAO implementing robust JDBC PreparedStatement routines for inserting appointment bookings, retrieving schedules by patient, and querying doctor daily queues. Implemented specialized isSlotBooked query routines and core MedicalRecordDAO persistence methods, validating relational integrity through extensive unit tests against our local MySQL instance. This verified complete readiness.
Problems Faced	SQL Time type mapping was confusing. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Created AppointmentService with input validation. (2) Added the double-booking check inside the service. (3) Implemented AppointmentServlet for booking requests.

Week 6 | 10-08-2026 to 16-08-2026 | Phase: Module coding - services and servlets

Week dropdown	Week 6
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Developed core application business services and HTTP servlet controllers managing appointment scheduling and clinical consultation lifecycles. Constructed AppointmentService incorporating strict server-side validation logic and automated double-booking prevention algorithms. Built AppointmentServlet to process incoming HTTP GET and POST requests for slot reservations, rescheduling adjustments, and cancellation actions. Integrated real-time appointment status transitions across Scheduled, Completed, and Cancelled states. Coupled servlet controllers with centralized audit logging to record timestamped transactional events across all user booking operations. This ensured dependable hospital system stability.
Problems Faced	Double-booking check allowed edge cases. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Created the patient book_appointment.jsp with doctor and slot selection. (2) Created the patient appointments.jsp with status badges. (3) Created the receptionist book_appointment.jsp.

Week 7 | 17-08-2026 to 23-08-2026 | Phase: Module coding - JSP views and UI

Week dropdown	Week 7
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Constructed responsive JavaServer Pages delivering interactive user interfaces for patients, receptionists, and clinical physicians. Developed patient book_appointment.jsp featuring dynamic physician selection and interactive time slot pickers. Created patient appointments.jsp displaying booking histories with color-coded status badges. Built receptionist book_appointment.jsp supporting front-desk scheduling and doctor appointments.jsp organizing daily patient queues chronologically. Engineered doctor consultation.jsp allowing physicians to record diagnostic observations, integrating common navigation headers, sidebars, and external style sheets to ensure consistent visual aesthetics across all module views. This worked reliably.
Problems Faced	Date picker accepted past dates. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Added past-date and empty reason validations. (2) Handled booking exceptions with friendly messages. (3) Added JavaScript to block past dates in the date picker.

Week 8 | 24-08-2026 to 30-08-2026 | Phase: Module coding - validation, exceptions, security

Week dropdown	Week 8
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Implemented rigorous server-side input validation, client-side date controls, and exception handling protocols across all appointment interfaces. Programmed validation routines rejecting past appointment dates, historical time slots, and blank consultation reasons. Implemented client-side JavaScript date pickers disabling previous calendar dates to prevent invalid submissions. Configured custom exception handlers displaying friendly diagnostic messages during double-booking conflicts. Restricted cancellation privileges strictly to booking owners and authorized receptionists, verifying security validation banners across all form templates. This established robust data consistency across hospital modules.
Problems Faced	Cancel was visible to wrong role. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Completed consultation saving into medical records. (2) Added the status change to Completed after consultation. (3) Added filters on appointment lists by date and status.

Week 9 | 31-08-2026 to 06-09-2026 | Phase: Module coding - module completion and self-review

Week dropdown	Week 9
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Finalized medical record persistence mechanisms and completed rigorous module refinement and code polishing prior to integration. Completed the consultation saving workflow in DoctorServlet, persisting clinical diagnostic notes and treatment recommendations directly into the medical_records database table. Added automated appointment status transitions updating records to Completed upon consultation closure. Implemented filtering mechanisms across appointment lists by date and status, conducted comprehensive source code reviews, inserted Javadoc comments, and pushed clean module code to the team repository. This established reliable module integration.
Problems Faced	Status did not update after consultation. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Integrated doctor list from Module 2 into booking. (2) Connected consultation with prescription and lab pages. (3) Fixed time format mismatch between JSP and DAO.

Week 10 | 07-09-2026 to 13-09-2026 | Phase: Integration of all five modules

Week dropdown	Week 10
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Coordinated multi-module integration linking appointment and consultation workflows with peer hospital subsystems. Integrated dynamic doctor directory feeds from Module 2 into appointment booking forms, allowing patients to filter available physicians by medical department. Connected consultation completion handlers with prescription and laboratory ordering views in Module 4. Resolved timestamp formatting discrepancies between JSP input controls and MySQL database fields. Verified unified views across patient, receptionist, and doctor roles, running comprehensive end-to-end booking through diagnostic workflows flawlessly. This established reliable module integration.
Problems Faced	Time format mismatch between JSP and DAO. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote test cases for booking, rescheduling and cancelling. (2) Tested double-booking prevention with same slot. (3) Tested consultation notes and record saving.

Week 11 | 14-09-2026 to 20-09-2026 | Phase: Testing (unit, functional, role-based)

Week dropdown	Week 11
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Executed comprehensive unit, functional, and role-based test suites across all appointment and consultation components. Formulated structured test cases evaluating booking, rescheduling, and cancellation workflows under diverse boundary conditions. Tested double-booking prevention routines by attempting concurrent reservations on identical physician time slots, confirming robust conflict rejection. Validated accurate clinical note persistence in medical records and verified role-based access restrictions on consultation interfaces. Logged identified defects, collaborated with team members to resolve bugs, and verified fixes through regression retesting. This worked reliably.
Problems Faced	A test case failed for same-time reschedule. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Tested booking flow on the live Render site. (2) Fixed time zone difference between local and cloud server. (3) Corrected status display bug on the appointments list.

Week 12 | 21-09-2026 to 27-09-2026 | Phase: Deployment (Docker, Render, cloud MySQL) and bug fixing

Week dropdown	Week 12
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Coordinated cloud deployment testing and resolved platform-specific scheduling discrepancies on the live production server. Verified all appointment booking and consultation JSP interfaces within the containerized web application running on Render. Identified and resolved a critical time zone disparity between local development clocks and cloud server environments affecting appointment timestamps. Corrected status badge rendering bugs on live appointment tables, retested consultation workflows on the production site, and updated deployment documentation with cloud configuration details. This guaranteed seamless clinical workflow operational efficiency.
Problems Faced	Time zone differed on cloud server. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote the appointment module section of the report. (2) Added flow diagrams and screenshots to the report. (3) Prepared slides on the appointment workflow.

Week 13 | 28-09-2026 to 04-10-2026 | Phase: Report, PPT, demo video and viva preparation

Week dropdown	Week 13
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Compiled technical documentation, recorded module demonstrations, and completed comprehensive viva preparation for academic evaluation. Authored the dedicated appointment and consultation module chapter of the final project report, detailing entity designs, double-booking algorithms, and clinical workflows. Formulated high-resolution sequence diagrams, UI screenshots, and relational table specifications. Designed presentation slides highlighting scheduling mechanics and medical record integration. Rehearsed live demonstrations of appointment booking and consultation workflows, preparing detailed technical answers on concurrency handling and slot validation. This ensured dependable hospital system stability.
Problems Faced	Exporting screenshots for report. Solved with team discussion and guide's suggestions.
Plan for Next Week	Final viva and demo, submit report, PPT, SRS, demo video and working software on the portal.

Part B - Daily Work Log (65 Entries) — Hansraj Pandey

Status for all entries: Completed. Saturday is the weekly mentor meeting, so no daily entry is listed.

Week 1 - Team formation, guide selection and abstract

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
06-07-2026 (Mon)	1.75	Completed	Participated in the initial project kickoff meeting with all five team members to evaluate hospital operational workflows ripe for digitization. We analyzed traditional clinical bottlenecks, including overcrowded outpatient waiting rooms, manual appointment paper registers, and lost patient visit records. We reached consensus on building the Smart Hospital Management System, mapping core appointment scheduling responsibilities.
07-07-2026 (Tue)	2	Completed	Conducted an empirical workflow study examining appointment booking and consultation management in local outpatient clinics. I observed how front-desk receptionists schedule patient visits, assign token numbers, and manage physician availability on paper charts. This field analysis highlighted severe scheduling inefficiencies, double-booking risks, and manual queue delays that our software must systematically resolve.
08-07-2026 (Wed)	2	Completed	Drafted the specialized abstract section addressing appointment scheduling and clinical consultation workflows for our SmartCare platform proposal. I documented core functional goals, including automated time-slot reservations, double-booking prevention, real-time doctor queues, and structured electronic diagnosis logging. I aligned the narrative with overarching project milestones, emphasizing how automated scheduling optimizes healthcare delivery.
09-07-2026 (Thu)	1.5	Completed	Consulted with our faculty project guide, Er. Ram Babu Buri, to review the technical architecture proposed for appointment management. We discussed using Java Servlets, JSP, and MySQL to implement robust transactional scheduling. Our guide emphasized concurrency control for shared doctor slots, thread safety in booking servlets, and normalized database designs for clinical records.
10-07-2026 (Fri)	1.75	Completed	Reviewed and refined the finalized project abstract alongside all module leads, verifying that appointment workflows integrated seamlessly with patient intake, medical prescribing, and billing modules. We incorporated our mentor's structural guidance, performed final proofreading, and assisted the team in successfully submitting the completed abstract proposal through the university project portal.

Week 2 - SRS (Software Requirement Specification)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
13-07-2026 (Mon)	1.75	Completed	Formulated detailed functional requirements for appointment scheduling within our formal Software Requirement Specification document. I specified comprehensive operational criteria for booking new clinic visits, selecting preferred doctors, rescheduling existing appointments, and processing patient cancellations. I established explicit rules for appointment confirmations, booking lead times, and daily schedule limits across hospital outpatient departments.
14-07-2026 (Tue)	2	Completed	Authored comprehensive functional requirements governing doctor consultations and clinical diagnostic records within the SRS document. I documented workflows enabling attending physicians to view daily patient queues, record clinical observations, document symptoms, enter provisional diagnoses, and recommend treatment plans. These specifications ensure standardized electronic medical records are captured consistently during every clinical visit.
15-07-2026 (Wed)	1.5	Completed	Defined the formal lifecycle states and status transitions for clinic appointments, establishing three core states: Scheduled, Completed, and Cancelled. I detailed the exact business conditions triggering state updates, such as receptionist check-in, physician consultation completion, or patient cancellation. This formal state model provided an unambiguous foundation for database schema design and business service logic.
16-07-2026 (Thu)	1.5	Completed	Engineered critical business logic rules and validation constraints within the specification to prevent appointment scheduling conflicts. I mandated strict double-booking prevention mechanisms ensuring that no physician can be scheduled for overlapping consultation time slots on the same date. Furthermore, I specified business rules governing cancellation deadlines and slot availability release mechanics.
17-07-2026 (Fri)	2	Completed	Consolidated all appointment scheduling and clinical consultation specifications into the master team Software Requirement Specification document. I led an internal review walkthrough with team members, ensuring clean integration with patient profiles from Module 2 and billing invoices from Module 5. We finalized requirement descriptions and received unanimous team approval on the SRS baseline.

Week 3 - UML design (use-case, class, sequence)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
20-07-2026 (Mon)	1.5	Completed	Engineered comprehensive Unified Modeling Language use-case diagrams for appointment scheduling and consultation workflows using Visual Paradigm. I mapped functional interactions for Patient, Doctor, and Receptionist actors, including booking appointments, viewing consultation schedules, rescheduling visits, and recording clinical notes. I established include and extend relationships representing mandatory validation checks and authentication prerequisites.
21-07-2026 (Tue)	2	Completed	Designed the structural class diagram for the core Appointment domain model entity in Java. I specified private attributes including appointmentId, patientId, doctorId, appointmentDate, timeSlot, status, and reasonForVisit. I defined complete accessor and mutator methods, constructor signatures, and relational associations linking appointments to patient profiles and attending doctor accounts. This ensured dependable hospital system stability.
22-07-2026 (Wed)	2	Completed	Constructed the structural class diagram for the MedicalRecord domain model, detailing its encapsulated attributes and associations. I defined fields for recordId, appointmentId, patientId, doctorId, symptoms, diagnosis, and visitDate. I established a strict one-to-one relationship linking each medical record to its corresponding completed appointment, ensuring clinical notes maintain permanent relational integrity.
23-07-2026 (Thu)	1.75	Completed	Drafted an intricate sequence diagram illustrating the dynamic runtime execution of an appointment booking request. The diagram details message flows between the patient web browser, AppointmentServlet controller, AppointmentService business layer, slot validation routines, and AppointmentDAO database persistence. I emphasized concurrency checks that inspect slot availability before confirming database transactions. This verified complete system reliability.
24-07-2026 (Fri)	1.5	Completed	Merged all Module 3 UML diagrams into the master project design document in collaboration with peer module leads. We harmonized naming conventions, verified foreign key references between doctor IDs and patient IDs, and eliminated structural redundancies across class diagrams. We confirmed that our integrated models satisfied all design standards outlined by our faculty mentor.

Week 4 - Database (ER) design, UI mock-ups, project setup

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
27-07-2026 (Mon)	2	Completed	Designed the normalized relational database schema for the appointments table in MySQL. I established columns for appointment_id, patient_id, doctor_id, appointment_date, time_slot, status, reason, and created_at timestamps. I configured primary key constraints, foreign key links to patients and doctors tables, and unique composite constraints preventing duplicate bookings on identical slots. This verified complete system reliability.
28-07-2026 (Tue)	1.75	Completed	Architected the relational schema for the medical_records table in MySQL to store clinical examination summaries. I defined columns for record_id, appointment_id, patient_id, doctor_id, symptoms, diagnosis, doctor_notes, and created_at timestamps. I established strict foreign key constraints linking records to appointments and patients, ensuring referential integrity across medical histories. This guaranteed seamless clinical workflow operational efficiency.
29-07-2026 (Wed)	1.5	Completed	Engineered strategic database indexes on the appointments table to optimize query performance during high-frequency front-desk lookups. I created composite indexes across doctor_id and appointment_date, as well as patient_id and appointment_date columns. These indexes dramatically reduce database search latency when fetching daily doctor queues and patient booking histories. This guaranteed seamless clinical workflow operational efficiency.
30-07-2026 (Thu)	1.75	Completed	Configured my local software development environment, installing Java Development Kit 17, Apache Tomcat 10.1, and MySQL Server 8.0. I cloned the shared team GitHub repository, imported the Maven web project into my development IDE, and verified that baseline servlet endpoints compile and deploy successfully to the local Tomcat runtime container.
31-07-2026 (Fri)	1.75	Completed	Crafted intuitive, responsive user interface wireframes and mock-ups for appointment booking forms and doctor consultation screens. Using modern layout tools, I designed booking interfaces featuring interactive date pickers, available time-slot selectors, and confirmation modals. For doctor consultation screens, I mapped out clean input panels for clinical observations and diagnostic notes.

Week 5 - Module coding - models and DAO layer

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
03-08-2026 (Mon)	2	Completed	Programmed the Appointment JavaBean model class in Java adhering strictly to enterprise encapsulation principles. I declared private instance variables for appointmentId, patientId, doctorId, appointmentDate, timeSlot, status, and reasonForVisit. I implemented full-parameter constructors, default constructors, public getters, setters, and overridden toString and equals methods to support reliable collection operations. This ensured dependable hospital system stability.
04-08-2026 (Tue)	1.5	Completed	Implemented the MedicalRecord domain model class in Java to represent clinical diagnosis records. I defined encapsulated fields for recordId, appointmentId, patientId, doctorId, symptoms, diagnosis, and visitDate. I authored clean accessor and mutator methods, ensuring seamless data exchange between web form parameters and backend database persistence layers during consultations. This ensured dependable hospital system stability.
05-08-2026 (Wed)	1.75	Completed	Constructed the AppointmentDAO interface and implemented core database persistence methods using JDBC. I developed insertAppointment and getAppointmentsByPatientId routines utilizing PreparedStatement to eliminate SQL injection vulnerabilities. I mapped ResultSet rows into instantiated Appointment objects, managing database connections safely using modern Java try-with-resources blocks to prevent resource leaks. This established robust data consistency across hospital modules.
06-08-2026 (Thu)	2	Completed	Expanded AppointmentDAO by programming advanced query routines, including getAppointmentsByDoctorAndDate and getAppointmentsByDate. These methods fetch daily doctor queues and departmental schedules using parameterized SQL statements. I tested these queries against our local MySQL database, confirming accurate data retrieval, proper date filtering, and optimal execution performance. This ensured dependable performance and verified complete hospital data integrity.
07-08-2026 (Fri)	1.5	Completed	Developed the isSlotBooked query method in AppointmentDAO and implemented baseline MedicalRecordDAO persistence routines. The isSlotBooked method executes high-performance checks verifying whether a doctor has an active booking for a given date and time slot. I verified query accuracy with standalone test scripts against our local database instance. This guaranteed seamless clinical workflow operational efficiency.

Week 6 - Module coding - services and servlets

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
10-08-2026 (Mon)	2	Completed	Engineered the AppointmentService business layer class in Java to encapsulate core scheduling logic and input validation rules. I implemented service methods to validate appointment dates, verify patient eligibility, and enforce business policies prior to database transactions. This service layer decouples HTTP servlet handling from data access routines, enhancing system maintainability.
11-08-2026 (Tue)	1.5	Completed	Implemented robust double-booking prevention algorithms within AppointmentService to resolve concurrent scheduling conflicts. The service invokes AppointmentDAO.isSlotBooked within a transactional boundary before executing booking insertions. If a requested doctor slot is already occupied, the service halts execution and throws a descriptive business exception, completely preventing duplicate reservations. This established robust data consistency across hospital modules.
12-08-2026 (Wed)	1.75	Completed	Developed the AppointmentServlet controller to process incoming HTTP GET and POST requests for appointment bookings. In the doGet handler, the servlet retrieves available doctors and forwards users to the booking view. In the doPost handler, it extracts form parameters, invokes AppointmentService, and redirects users with success or error notifications. This verified complete system reliability.
13-08-2026 (Thu)	2	Completed	Expanded AppointmentServlet by implementing controller actions for rescheduling and cancelling clinic appointments. I programmed request dispatchers that handle date-time updates and status modifications, ensuring that cancelled slots are released immediately for other patients. I added validation checks verifying that past appointments cannot be rescheduled or cancelled by users. This ensured dependable hospital system stability.
14-08-2026 (Fri)	2	Completed	Integrated automated status transitions and audit logging mechanisms into AppointmentServlet controller actions. I programmed logic that records status changes from Scheduled to Cancelled or Completed in the database. Furthermore, I integrated AuditDAO calls capturing timestamped user actions, IP addresses, and appointment identifiers for comprehensive administrative traceability across all booking events.

Week 7 - Module coding - JSP views and UI

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
17-08-2026 (Mon)	2	Completed	Developed the patient-facing book_appointment.jsp view using semantic HTML5 and responsive CSS flexbox styling. I designed an intuitive booking interface featuring dynamic doctor selection dropdowns, interactive date pickers, and organized time-slot grid buttons. I integrated client-side controls that update available time slots dynamically when users choose different physicians or dates. This verified complete system reliability.
18-08-2026 (Tue)	1.5	Completed	Constructed the patient appointments.jsp interface allowing registered patients to inspect their comprehensive booking histories. I built a clean, responsive table displaying appointment dates, assigned doctors, clinical departments, time slots, and reasons for visit. I integrated color-coded JSTL status badges (Scheduled, Completed, Cancelled) to provide immediate visual clarity. This guaranteed seamless clinical workflow operational efficiency.
19-08-2026 (Wed)	2	Completed	Engineered the receptionist book_appointment.jsp interface to streamline front-desk patient scheduling operations. I designed an efficient, high-speed booking form enabling receptionists to search registered patients by ID or phone number, assign doctors, and reserve immediate consultation slots. The interface provides instant feedback, confirming appointments without reloading entire page layouts. This ensured dependable hospital system stability.
20-08-2026 (Thu)	2	Completed	Built the doctor appointments.jsp interface delivering a real-time clinical queue view for hospital physicians. I developed a structured tabular view listing the day's scheduled patients chronologically by appointment time. I incorporated quick-action buttons allowing doctors to mark patient arrivals and launch clinical consultation screens directly from the queue. This ensured dependable hospital system stability.
21-08-2026 (Fri)	1.75	Completed	Developed the doctor consultation.jsp view allowing attending physicians to record clinical findings during patient examinations. I created dedicated form sections for entering clinical symptoms, vital observations, provisional diagnoses, and recommended treatments. I integrated common navigation headers, sidebars, and external style sheets, ensuring visual and structural consistency across all screens. This verified complete system reliability.

Week 8 - Module coding - validation, exceptions, security

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
24-08-2026 (Mon)	2	Completed	Implemented rigorous server-side validation rules in AppointmentServlet to eliminate invalid booking payloads. I authored validation checks verifying that appointment dates are in the future, selected time slots match standardized hospital consultation intervals, and reason-for-visit text fields are non-empty. Form submissions failing validation are returned immediately with field-specific guidance messages. This verified complete system reliability.
25-08-2026 (Tue)	1.75	Completed	Constructed comprehensive exception handling logic across booking servlets to manage scheduling conflicts gracefully. I caught double-booking exceptions thrown by AppointmentService, mapping them to friendly, localized warning messages on the JSP view. This ensures patients receive clear recommendations to select alternative time slots rather than encountering unhandled server 500 errors. This verified complete system reliability.
26-08-2026 (Wed)	2	Completed	Developed client-side JavaScript date validation scripts to enhance user experience on booking forms. I added logic that sets the minimum selectable date on HTML5 calendar inputs to today's date, preventing users from picking past dates. This instant client-side barrier prevents invalid HTTP submissions and reduces unnecessary server-side validation overhead. This verified complete system reliability.
27-08-2026 (Thu)	2	Completed	Implemented strict role-based access control and authorization checks on appointment cancellation actions. I programmed servlet logic verifying that an incoming cancellation request originates either from the patient who owns the booking or an authorized receptionist. If an unauthorized user attempts cancellation, the servlet rejects the action, returning an HTTP 403 Forbidden error.
28-08-2026 (Fri)	1.5	Completed	Conducted comprehensive validation and error handling testing across all Module 3 form interfaces. I submitted boundary test inputs, including empty forms, malformed dates, script tags, and duplicate bookings. I verified that validation error banners render cleanly, form inputs retain entered values upon failure, and unauthorized access attempts are blocked. This verified complete system reliability.

Week 9 - Module coding - module completion and self-review

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
31-08-2026 (Mon)	1.75	Completed	Completed the clinical consultation persistence pipeline within DoctorServlet, saving diagnostic observations into the database. When a physician submits the consultation.jsp form, the servlet extracts clinical notes, invokes MedicalRecordDAO to persist records, and links them to the active appointment ID, creating a permanent electronic health record of the visit. This ensured dependable hospital system stability.
01-09-2026 (Tue)	1.5	Completed	Implemented automated appointment status update logic that triggers upon consultation submission. In the same database transaction where medical records are saved, the servlet updates the appointment status from Scheduled to Completed. This automated transition ensures daily doctor queues update in real time, preventing duplicate consultation entries for completed visits. This verified complete system reliability.
02-09-2026 (Wed)	1.5	Completed	Engineered dynamic search and filtering controls across appointment listing tables in both patient and administrative views. I implemented SQL queries and servlet parameters allowing users to filter bookings by specific date ranges, attending doctors, and status categories. These filters allow front-desk staff to audit daily schedules efficiently without manual paging.
03-09-2026 (Thu)	1.75	Completed	Conducted a comprehensive code review and cleanup across all Module 3 Java packages. I eliminated redundant methods, optimized SQL query strings, and standardized variable naming conventions across DAOs and servlets. Furthermore, I authored detailed Javadoc comments on all public interfaces and service methods, documenting parameters, return values, and thrown exceptions.
04-09-2026 (Fri)	1.75	Completed	Committed the finalized, documented Module 3 codebase to our shared team GitHub repository following standard Git branch workflows. I updated our team project tracking spreadsheet, detailing completed servlet endpoints, DAO queries, and JSP views. I coordinated with peer module leads to confirm readiness for next week's system-wide integration. This ensured dependable hospital system stability.

Week 10 - Integration of all five modules

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
07-09-2026 (Mon)	1.5	Completed	Integrated the active doctor directory from Module 2 into the appointment booking workflow. I modified AppointmentServlet to query DoctorDAO dynamically, populating booking forms with certified physicians categorized by department and specialization. This enables patients to choose specific doctors and view accurate consultation fees during the booking process without manual data entry.
08-09-2026 (Tue)	2	Completed	Connected the consultation completion workflow with prescription and laboratory ordering interfaces in Module 4. When a doctor completes a consultation, the system provides direct action buttons to prescribe medications and request lab tests for the patient, pre-populating patient identifiers and consultation references across modules to streamline clinical handoffs. This ensured dependable hospital system stability.
09-09-2026 (Wed)	1.75	Completed	Investigated and resolved a critical time format mismatch between JavaServer Pages input fields and MySQL TIME datatypes. The web form submitted twelve-hour AM/PM timestamps, whereas the database expected standard twenty-four-hour formats. I implemented format conversion utilities in AppointmentServlet, ensuring reliable date-time parsing and error-free database persistence. This established robust data consistency across hospital modules.
10-09-2026 (Thu)	1.75	Completed	Verified unified multi-role views across patient, receptionist, and doctor interfaces for individual appointment records. I confirmed that when a receptionist schedules an appointment, it immediately appears on the patient's portal, updates the doctor's daily queue, and reflects accurate status badges across all three dashboards without synchronization delays or cache inconsistencies.
11-09-2026 (Fri)	2	Completed	Executed comprehensive end-to-end integration walkthroughs validating the entire clinical appointment lifecycle. I simulated booking an appointment as a patient, checking in through reception, completing the examination as a doctor, and recording clinical notes into medical records. All transitions functioned reliably, confirming seamless data flow across integrated application modules. This ensured dependable hospital system stability.

Week 11 - Testing (unit, functional, role-based)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
14-09-2026 (Mon)	2	Completed	Designed and executed comprehensive unit and functional test suites evaluating appointment booking, rescheduling, and cancellation routines. I formulated test cases covering standard bookings, date-time updates, and cancellations under various clinical conditions, verifying that all database transactions execute reliably and persist updated appointment states accurately without data loss. This guaranteed seamless clinical workflow operational efficiency.
15-09-2026 (Tue)	1.75	Completed	Conducted rigorous concurrency and double-booking prevention testing on the slot reservation subsystem. I simulated simultaneous booking attempts for the same doctor and time slot across multiple browser sessions. I confirmed that AppointmentService intercepts conflicting requests, allowing only the first transaction to persist while rejecting subsequent attempts with clear warnings. This verified complete system reliability.
16-09-2026 (Wed)	1.75	Completed	Tested medical record persistence and diagnostic note accuracy under diverse consultation scenarios. I submitted clinical notes containing complex medical terminology, formatting characters, and multiline text. I verified that MedicalRecordDAO persists data without character truncation, and validated that past clinical records display accurately on patient history timelines. This established robust data consistency across hospital modules.
17-09-2026 (Thu)	2	Completed	Executed role-based security testing across all appointment and consultation URL endpoints. I verified that patients attempting to access doctor consultation servlets directly are blocked with HTTP 403 Forbidden responses. Similarly, I confirmed that doctors cannot cancel other physicians' bookings, validating that AuthorizationFilter enforces strict role segregation. This established robust data consistency across hospital modules.
18-09-2026 (Fri)	1.75	Completed	Logged and documented all defects uncovered during testing in our centralized bug tracking log. I collaborated with team members to resolve high-priority issues, including fixing a same-time reschedule glitch and correcting a status badge display anomaly. We retested all modified components to ensure complete regression stability across the platform. This verified complete system reliability.

Week 12 - Deployment (Docker, Render, cloud MySQL) and bug fixing

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
21-09-2026 (Mon)	1.75	Completed	Performed live deployment testing of the appointment and consultation subsystem on the Render cloud hosting platform. I accessed the live public web application, scheduled multiple test appointments, and verified that containerized Tomcat instances execute booking servlets and serve dynamic JSP views reliably without connection timeouts or runtime exceptions. This ensured dependable hospital system stability.
22-09-2026 (Tue)	1.5	Completed	Investigated and resolved a critical time zone disparity between local development workstations and the cloud server. The cloud container operated on UTC time, causing scheduled appointment times to display with hour offsets on local browsers. I configured JVM time zone parameters in the Dockerfile to Asia/Kolkata, ensuring synchronized date-time rendering.
23-09-2026 (Wed)	1.75	Completed	Corrected a minor status badge rendering bug on live appointment listing tables. On the cloud database, certain legacy test records with mixed-case status values caused CSS badge classes to fail matching. I standardized status string comparisons using case-insensitive equals in JSP JSTL tags, restoring clean, color-coded status badges across all browsers.
24-09-2026 (Thu)	2	Completed	Retested the complete doctor consultation workflow directly on the live cloud production environment. I logged in as an attending physician, launched the consultation interface, recorded diagnostic findings for test patients, and verified immediate persistence in cloud MySQL. The entire workflow executed smoothly, confirming production readiness for live evaluations. This ensured dependable hospital system stability.
25-09-2026 (Fri)	1.75	Completed	Compiled detailed deployment notes documenting cloud configuration settings, time zone synchronizations, and database connection parameters. I updated the deployment chapter of our team project report, cataloging environmental differences between local and cloud tiers. I verified that all Module 3 features operate reliably on the live public URL. This guaranteed seamless clinical workflow operational efficiency.

Week 13 - Report, PPT, demo video and viva preparation

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
28-09-2026 (Mon)	1.5	Completed	Authored the complete Appointment and Consultation chapter of our team's final Project Based Learning technical report. I documented module architecture, domain entity relationships, double-booking prevention algorithms, and servlet dispatching workflows. I explained how our design ensures high concurrency safety, data normalization, and seamless clinical handoffs throughout outpatient hospital operations. This verified complete system reliability.
29-09-2026 (Tue)	2	Completed	Constructed detailed operational flow diagrams, sequence schematics, and high-resolution user interface screenshots for inclusion in the final report. I illustrated appointment booking sequences, conflict detection logic, and doctor consultation interfaces. I added descriptive captions and aligned figure references with report narrative to present a clear, scholarly deliverable. This guaranteed seamless clinical workflow operational efficiency.
30-09-2026 (Wed)	2	Completed	Designed professional, high-impact presentation slides summarizing Module 3 capabilities for our upcoming academic project defense. I structured slide decks covering clinical problem statements, slot reservation algorithms, database entity relationships, and live system screenshots. I rehearsed presentation timing, ensuring our scheduling highlights fit smoothly within the allotted faculty evaluation window. This verified complete system reliability.
01-10-2026 (Thu)	1.75	Completed	Rehearsed live system demonstration workflows showcasing appointment booking, front-desk intake, and doctor consultation recording. I practiced seamless handoffs between patient slot reservations and doctor diagnostic note entries, timing the presentation to fit within the allotted evaluation window. I verified that all demonstration accounts functioned reliably on the cloud server. This verified complete system reliability.
02-10-2026 (Fri)	2	Completed	Completed rigorous viva voce preparation focusing on core Java enterprise concepts, JDBC transaction management, and service-layer validation patterns. I rehearsed technical explanations regarding double-booking conflict resolution, database indexing strategies, and time zone synchronization. I organized project source files, ensuring complete readiness for final evaluation during the viva defense. This ensured dependable hospital system stability.

Total hours logged for Hansraj Pandey: 116.25 hours over 65 days.

4. Aryan Daware (24E1ARADM30P020)

Module 4 - Pharmacy, Laboratory and Nursing

Member	Aryan Daware (24E1ARADM30P020)
Module	Module 4 - Pharmacy, Laboratory and Nursing
Code owned	Medicine, MedicineInventory, Prescription, PrescriptionItem, LabTest, PatientVital models and DAOs, MedicineServlet (/inventory), PrescriptionServlet, LabServlet, NurseServlet, inventory, prescription_view, lab_management, lab_test_details, nurse JSPs.
Period	06-07-2026 to 04-10-2026 (13 weeks). Daily hours between 1.5 and 2.0.
GitHub link	https://github.com/mathurdevansh/Smart-Hospital-Management-System

Part A - Weekly Reports (13 Weeks) — Aryan Daware

Week 1 | 06-07-2026 to 12-07-2026 | Phase: Team formation, guide selection and abstract

Week dropdown	Week 1
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Collaborated with project team members during initial kickoff meetings to define functional boundaries for the Pharmacy, Laboratory, and Nursing subsystem. Researched clinical workflows in community healthcare facilities, focusing on pharmaceutical stock maintenance, diagnostic specimen tracking, and nurse vital sign observation charts. Authored the specialized abstract section detailing electronic prescriptions, laboratory test result lifecycles, and inpatient vital sign monitoring. Consulted with our faculty mentor and teammates to validate the Java Servlet-JSP and MySQL architecture, subsequently proofreading and submitting the unified abstract proposal to the.
Problems Faced	Too many ideas to compare. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Listed requirements for medicine stock and expiry tracking. (2) Listed requirements for doctor prescriptions with medicine items. (3) Listed requirements for lab test request and result entry.

Week 2 | 13-07-2026 to 19-07-2026 | Phase: SRS (Software Requirement Specification)

Week dropdown	Week 2
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Formulated comprehensive functional and non-functional requirements for Module 4 within the team Software Requirement Specification. Cataloged detailed operational criteria for pharmacy inventory tracking, batch expiry management, multi-item doctor prescriptions, laboratory diagnostic test requests, and nurse vital sign recordings. Specified stringent business logic rules preventing expired medication dispensing and invalid physiological measurement entries. Documented non-functional standards addressing transactional consistency, database throughput, and user interface responsiveness. Consolidated all documentation into the master SRS and conducted a thorough architectural walkthrough with teammates to verify completeness.
Problems Faced	Many pharmacy rules to capture. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Drew the use-case diagram for Nurse and Lab actors. (2) Designed the class diagram for Medicine and MedicineInventory. (3) Designed the class diagram for Prescription and PrescriptionItem.

Week 3 | 20-07-2026 to 26-07-2026 | Phase: UML design (use-case, class, sequence)

Week dropdown	Week 3
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Engineered comprehensive Unified Modeling Language diagrams mapping the structural and behavioral architecture of the pharmacy, laboratory, and nursing subsystem. Formulated detailed use-case diagrams modeling operational boundaries for Nurse and Lab Technician actors. Designed structural class diagrams for Medicine, MedicineInventory, Prescription, PrescriptionItem, LabTest, and PatientVital domain entities, detailing encapsulated attributes, constructor definitions, and operational methods. Constructed dynamic sequence diagrams detailing multi-tier prescription dispensing and laboratory result entry workflows. Merged all module diagrams into the master project design document following mentor guidelines.
Problems Faced	Parent-child relation for prescription items. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Designed the medicines and medicine_inventory tables. (2) Designed prescriptions and prescription_items tables. (3) Designed lab_tests and patient_vitals tables.

Week 4 | 27-07-2026 to 02-08-2026 | Phase: Database (ER) design, UI mock-ups, project setup

Week dropdown	Week 4
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Architected normalized relational database schemas and established the local engineering environment for Module 4. Designed normalized MySQL tables for medicines, medicine_inventory, prescriptions, prescription_items, lab_tests, and patient_vitals with strict primary and foreign key constraints. Set up our local development environment with JDK 17, Apache Tomcat 10.1, and MySQL Server 8.0, successfully cloning the shared team GitHub repository. Created high-fidelity user interface wireframes and mock-ups for pharmacy inventory management tables, doctor prescription writing pads, and nursing clinical vitals recording screens.
Problems Faced	Foreign key order in schema. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote Medicine and MedicineInventory model classes. (2) Wrote Prescription and PrescriptionItem models. (3) Wrote LabTest and PatientVital models.

Week 5 | 03-08-2026 to 09-08-2026 | Phase: Module coding - models and DAO layer

Week dropdown	Week 5
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Implemented foundational data access layer components and domain models for pharmacy inventory, electronic prescriptions, laboratory investigations, and patient vitals. Authored Medicine, MedicineInventory, Prescription, PrescriptionItem, LabTest, and PatientVital Java JavaBean model classes adhering strictly to encapsulation standards. Constructed MedicineDAO providing robust inventory stock queries and batch tracking methods. Implemented PrescriptionDAO, LabDAO, and VitalDAO featuring secure JDBC PreparedStatement routines for transactional data persistence, validating relational integrity through extensive unit tests against our local MySQL test database. This ensured dependable hospital system stability.
Problems Faced	Mapping decimal fields correctly. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Implemented MedicineServlet for inventory add and update. (2) Implemented PrescriptionServlet to save multiple medicine items. (3) Implemented LabServlet for test request and result update.

Week 6 | 10-08-2026 to 16-08-2026 | Phase: Module coding - services and servlets

Week dropdown	Week 6
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Developed core application business services and HTTP servlet controllers managing pharmacy stock, electronic prescriptions, laboratory diagnostic tests, and nursing vitals. Built MedicineServlet handling inventory replenishment and batch updates. Implemented PrescriptionServlet supporting atomic multi-item medication orders within transactional boundaries. Developed LabServlet managing diagnostic test requests and technician result submissions. Constructed NurseServlet to record and display patient vital sign trends. Integrated automated low-stock threshold triggers and medication expiry evaluation algorithms within the underlying data access layer. This ensured dependable hospital system stability.
Problems Faced	Saving many items needed a loop with transaction. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Created the inventory.jsp with stock table and add form. (2) Created prescriptions.jsp for the doctor and prescription_view.jsp. (3) Created lab_management.jsp and lab_test_details.jsp.

Week 7 | 17-08-2026 to 23-08-2026 | Phase: Module coding - JSP views and UI

Week dropdown	Week 7
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Constructed responsive JavaServer Pages delivering interactive user interfaces for pharmacists, laboratory technicians, clinical nurses, and medical doctors. Built inventory.jsp featuring searchable stock tables and batch entry forms. Developed doctor prescriptions.jsp and patient prescription_view.jsp interfaces for medication authoring and inspection. Engineered lab_management.jsp and lab_test_details.jsp providing laboratory technicians with diagnostic entry consoles. Constructed nurse_dashboard.jsp, nurse patients.jsp, record_vitals.jsp, and vitals_view.jsp, integrating common navigation headers, sidebars, and external style sheets to ensure consistent visual aesthetics across all module views. This established reliable module integration.
Problems Faced	Dynamic rows in prescription form. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Added validation for quantity, dose and vitals ranges. (2) Handled exceptions for stock below zero cases. (3) Added JavaScript validation to vitals form.

Week 8 | 24-08-2026 to 30-08-2026 | Phase: Module coding - validation, exceptions, security

Week dropdown	Week 8
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Implemented rigorous server-side input validation, client-side scripting controls, and exception handling protocols across all pharmacy, laboratory, and nursing interfaces. Added validation logic verifying medication quantities, dosage frequencies, and physiological vital sign ranges. Handled inventory stock-out exceptions gracefully, preventing negative inventory balances through transactional rollbacks. Developed client-side JavaScript validation scripts providing real-time feedback on clinical vitals forms. Implemented role-based URL protection restricting laboratory and pharmacy consoles, verifying validation banners across all form templates. This established robust data consistency across hospital modules.
Problems Faced	Stock could go negative. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Reduced stock automatically when a prescription is issued. (2) Added lab result status flow (Requested to Completed). (3) Added vital history list for each patient.

Week 9 | 31-08-2026 to 06-09-2026 | Phase: Module coding - module completion and self-review

Week dropdown	Week 9
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Finalized automated inventory deduction mechanics and completed rigorous module refinement and code polishing prior to integration. Implemented automated stock reduction algorithms that deduct batch quantities immediately upon prescription issuance. Developed complete laboratory result lifecycle states from Requested through Sample Collected to Completed. Engineered longitudinal vital sign historical timelines for patient profiles. Conducted comprehensive source code reviews, refactored data access methods, inserted descriptive Javadoc comments, and pushed clean module code to the shared GitHub repository. This ensured dependable hospital system stability.
Problems Faced	Lab status flow had gaps. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Linked prescriptions with consultation from Module 3. (2) Linked lab tests to the doctor's lab_reports page. (3) Linked patient prescriptions page with prescription data.

Week 10 | 07-09-2026 to 13-09-2026 | Phase: Integration of all five modules

Week dropdown	Week 10
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Participated actively in the multi-module system integration, linking pharmacy, laboratory, and nursing workflows with peer hospital subsystems. Integrated doctor prescription creation directly with clinical consultation completion screens from Module 3. Connected diagnostic test requests to the attending physician's clinical review console. Linked patient prescription history views with live pharmacy dispensing data. Resolved foreign key constraint conflicts identified during cross-module data insertion testing, and executed comprehensive end-to-end user journeys from doctor consultation to medication dispensing. This ensured dependable hospital system stability.
Problems Faced	Foreign key errors after merge. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote test cases for inventory and prescription. (2) Tested stock deduction with multiple medicines. (3) Tested lab request and result entry.

Week 11 | 14-09-2026 to 20-09-2026 | Phase: Testing (unit, functional, role-based)

Week dropdown	Week 11
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Executed comprehensive unit, functional, and role-based test suites across all pharmacy, laboratory, and nursing components. Formulated structured test cases evaluating inventory management, multi-item prescription processing, laboratory test status updates, and nurse vitals logging under diverse clinical conditions. Tested automated stock deductions under concurrent prescription submissions, validating atomic database transactions. Evaluated vital sign range validation against extreme boundary values, logged identified software defects, and verified fixes through exhaustive regression retesting cycles. This ensured dependable performance and verified complete hospital data integrity.
Problems Faced	Edge case in stock deduction. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Checked inventory and prescription pages on the live site. (2) Fixed foreign key insert order in seed data. (3) Corrected date display in the lab details page.

Week 12 | 21-09-2026 to 27-09-2026 | Phase: Deployment (Docker, Render, cloud MySQL) and bug fixing

Week dropdown	Week 12
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Coordinated cloud deployment testing and resolved platform-specific database discrepancies on the live production server. Verified all pharmacy, laboratory, and nursing JSP interfaces and servlet controllers within the Dockerized container running on Render. Identified and corrected foreign key insertion ordering issues within cloud database seed scripts. Fixed date formatting anomalies on laboratory test result detail pages, retested nurse vitals recording on the live server, and cataloged deployment configuration notes in the project report. This established robust data consistency across hospital modules.
Problems Faced	Seed data failed on cloud DB. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote the pharmacy, lab and nursing section of the report. (2) Added screenshots of inventory, prescription and vitals. (3) Prepared slides on pharmacy and lab workflow.

Week 13 | 28-09-2026 to 04-10-2026 | Phase: Report, PPT, demo video and viva preparation

Week dropdown	Week 13
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Compiled technical documentation, recorded module demonstrations, and completed comprehensive viva preparation for academic evaluation. Authored the dedicated pharmacy, laboratory, and nursing chapter of the final project report, detailing entity designs, inventory algorithms, and clinical workflows. Formulated high-resolution schema diagrams, UI screenshots, and relational table specifications. Designed presentation slides highlighting pharmaceutical stock workflows and laboratory lifecycles. Rehearsed live system demonstrations of prescription dispensing, and prepared detailed technical answers on database transactions and foreign key architectures. This ensured dependable hospital system stability.
Problems Faced	Many screenshots to collect. Solved with team discussion and guide's suggestions.
Plan for Next Week	Final viva and demo, submit report, PPT, SRS, demo video and working software on the portal.

Part B - Daily Work Log (65 Entries) — Aryan Daware

Status for all entries: Completed. Saturday is the weekly mentor meeting, so no daily entry is listed.

Week 1 - Team formation, guide selection and abstract

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
06-07-2026 (Mon)	1.75	Completed	Participated in the project kickoff meeting with all five team members to evaluate hospital operational workflows ripe for digitization. We analyzed traditional clinical bottlenecks in hospital pharmacies and diagnostic laboratories, including manual medicine stock ledgers, illegible handwritten prescriptions, and lost diagnostic lab reports. We established collaborative development standards and assigned individual module ownership.
07-07-2026 (Tue)	1.5	Completed	Conducted an empirical workflow study examining pharmacy dispensing and laboratory testing procedures across local community healthcare centers. I documented standard inventory replenishment routines, batch expiry tracking practices, physician prescription orders, and nursing vitals recording logs. This operational analysis established foundational data requirements for automating pharmaceutical stock control and diagnostic laboratory management.
08-07-2026 (Wed)	1.5	Completed	Drafted the specialized abstract section addressing pharmacy inventory, electronic prescriptions, laboratory test management, and nursing vitals for our SmartCare platform proposal. I documented core functional goals, including automated stock deduction, electronic prescription items, diagnostic test lifecycles, and physiological vital sign tracking, aligning the narrative with overarching project milestones. This ensured dependable hospital system stability.
09-07-2026 (Thu)	1.75	Completed	Met with our faculty project guide, Er. Ram Babu Buri, to evaluate proposed technology architectures against department research standards. We discussed using Java Servlets, JSP, and MySQL to implement transactional inventory updates and multi-item prescription management. Our guide emphasized ACID compliance during stock deduction and normalized database designs for clinical tests.
10-07-2026 (Fri)	1.75	Completed	Reviewed and refined the finalized project abstract alongside all module leads, verifying that pharmacy, lab, and nursing workflows integrated cleanly with patient registration, doctor appointments, and billing modules. We incorporated our mentor's structural guidance, performed final proofreading, and assisted the team in successfully submitting the completed abstract proposal through the university portal.

Week 2 - SRS (Software Requirement Specification)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
13-07-2026 (Mon)	2	Completed	Formulated detailed functional requirements for pharmacy stock inventory and medication expiry tracking within our formal Software Requirement Specification document. I specified comprehensive operational criteria for cataloging pharmaceutical drugs, managing reorder thresholds, tracking manufacturer batch numbers, and monitoring expiration dates to prevent dispensing expired pharmaceuticals to patients across the hospital facility.
14-07-2026 (Tue)	1.75	Completed	Authored comprehensive functional requirements governing doctor prescription authoring within the SRS document. I documented specifications allowing licensed physicians to issue multi-item electronic prescriptions linked to patient consultations. I detailed required fields for medication names, dosage quantities, intake frequencies, duration instructions, and special dietary precautions to ensure patient treatment safety. This verified complete system reliability.
15-07-2026 (Wed)	1.5	Completed	Documented functional requirements for laboratory diagnostic test requests and technician result entry workflows within the specification document. I cataloged requirements for requesting blood tests, radiology scans, and pathology screenings. I defined operational states for sample collection, diagnostic result entry, normal reference range comparisons, and attending physician report delivery. This ensured dependable hospital system stability.
16-07-2026 (Thu)	2	Completed	Specified comprehensive functional requirements for clinical nurse vital sign observations within the SRS document. I detailed specifications for recording physiological parameters, including systolic and diastolic blood pressure, pulse rate, body temperature, respiratory rate, and oxygen saturation. These requirements ensure inpatient monitoring charts capture accurate, timestamped patient physiological trends. This ensured dependable hospital system stability.
17-07-2026 (Fri)	2	Completed	Consolidated the pharmacy, laboratory, and nursing specifications into the master team Software Requirement Specification document. I led an internal review walkthrough with team members, ensuring clean integration with patient records from Module 2 and billing invoices from Module 5. We finalized requirement descriptions and received unanimous team approval on the SRS baseline.

Week 3 - UML design (use-case, class, sequence)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
20-07-2026 (Mon)	1.75	Completed	Engineered comprehensive Unified Modeling Language use-case diagrams for Nurse and Laboratory Technician actors using Visual Paradigm. I mapped functional interactions including dispensing medications, logging incoming pharmaceutical stock, updating diagnostic test results, and recording inpatient vital signs. I established include and extend relationships representing mandatory validation checks and prerequisite authentication controls.
21-07-2026 (Tue)	1.75	Completed	Designed the structural class diagram for the Medicine and MedicineInventory domain models in Java. I specified private attributes for medicineId, brandName, genericName, manufacturer, unitPrice, batchNumber, stockQuantity, and expiryDate. I defined complete accessor and mutator methods, constructor signatures, and relational associations linking pharmaceutical catalog items to physical stock inventory records. This verified complete system reliability.
22-07-2026 (Wed)	1.75	Completed	Constructed detailed structural class diagrams for the Prescription and PrescriptionItem domain entities, establishing a parent-child relationship. I defined fields for prescriptionId, consultationId, patientId, doctorId, issueDate, medicineId, dosage, frequency, and duration. This composite class structure ensures that multi-drug prescription orders maintain strict relational integrity in application memory. This established robust data consistency across hospital modules.
23-07-2026 (Thu)	1.75	Completed	Designed the structural class diagrams for LabTest and PatientVital domain entities, modeling clinical diagnostic investigations and physiological observations. I defined attributes for testId, testName, testCategory, testResult, normalRange, vitalId, systolicBP, diastolicBP, heartRate, and temperature. I established relational linkages connecting clinical test records to patient medical profiles. This verified complete operational readiness and high system maintainability.
24-07-2026 (Fri)	1.75	Completed	Drafted an intricate sequence diagram modeling the dynamic runtime execution of an electronic prescription issuance workflow. The diagram details message flows between the doctor web browser, PrescriptionServlet controller, inventory validation routines, and PrescriptionDAO database persistence. I merged all Module 4 diagrams into the master project design document following mentor guidelines.

Week 4 - Database (ER) design, UI mock-ups, project setup

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
27-07-2026 (Mon)	1.5	Completed	Designed the normalized relational database schemas for the medicines and medicine_inventory tables in MySQL. I established columns for medicine_id, brand_name, generic_name, dosage_form, unit_price, batch_number, stock_quantity, and expiry_date. I configured primary key constraints, unique barcode indexes, and foreign key linkages ensuring that inventory batches reference valid catalog items. This guaranteed seamless clinical workflow operational efficiency.
28-07-2026 (Tue)	1.75	Completed	Architected the relational schemas for prescriptions and prescription_items tables in MySQL to store multi-item medical orders. I structured columns for prescription_id, patient_id, doctor_id, consultation_id, issue_date, medicine_id, dosage, frequency, and quantity. I established cascading foreign key constraints linking prescription line items to parent prescriptions and pharmaceutical inventory records. This guaranteed seamless clinical workflow operational efficiency.
29-07-2026 (Wed)	1.75	Completed	Designed the relational database schemas for the lab_tests and patient_vitals tables in MySQL. I defined columns for test_id, patient_id, doctor_id, test_name, sample_type, result_value, normal_range, test_status, vital_id, temperature, blood_pressure, pulse_rate, and recorded_time. I configured foreign key constraints linking clinical records directly to registered patient profiles. This ensured dependable performance and verified complete hospital data integrity.
30-07-2026 (Thu)	1.75	Completed	Configured my local software development environment, installing Java Development Kit 17, Apache Tomcat 10.1, and MySQL Server 8.0. I cloned the shared team GitHub repository, imported the Maven web project into my development IDE, and verified that baseline servlet endpoints compile and deploy successfully to the local Tomcat runtime container.
31-07-2026 (Fri)	2	Completed	Crafted intuitive, responsive user interface wireframes and mock-ups for pharmacy inventory management, doctor prescription writing pads, and nursing vitals recording forms. Using modern layout tools, I designed responsive interfaces featuring searchable medicine tables, dynamic prescription line-item rows, and color-coded vital sign input gauges for rapid clinical data entry. This ensured dependable hospital system stability.

Week 5 - Module coding - models and DAO layer

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
03-08-2026 (Mon)	1.75	Completed	Programmed the Medicine and MedicineInventory JavaBean domain model classes in Java adhering strictly to enterprise encapsulation principles. I declared private instance variables for all drug catalog and inventory batch attributes, providing comprehensive parameterized and default constructors. I implemented standard getter and setter methods, alongside overridden toString, equals, and hashCode methods.
04-08-2026 (Tue)	1.75	Completed	Implemented the Prescription and PrescriptionItem domain model classes in Java to represent composite medical prescription orders. I encapsulated fields for prescription headers and detailed medication line items, including dosage instructions and quantities. I authored clean accessor and mutator methods, ensuring seamless data exchange between web form parameters and backend persistence layers.
05-08-2026 (Wed)	1.5	Completed	Constructed the LabTest and PatientVital domain model classes in Java to represent diagnostic investigations and physiological vital sign observations. I encapsulated fields for laboratory test categories, diagnostic results, normal reference ranges, and vital measurements. I implemented complete constructors and property accessors, supporting reliable collection operations across clinical services. This ensured dependable hospital system stability.
06-08-2026 (Thu)	1.75	Completed	Implemented the MedicineDAO interface and developed core database persistence methods using JDBC. I developed addMedicine, updateStockQuantity, and getMedicinesInStock routines utilizing PreparedStatement to eliminate SQL injection vulnerabilities. I mapped ResultSet rows into instantiated Medicine objects, managing database connections safely using modern Java try-with-resources blocks. This achieved excellent operational throughput and preserved data integrity across records.
07-08-2026 (Fri)	1.5	Completed	Constructed PrescriptionDAO, LabDAO, and VitalDAO classes featuring robust JDBC routines for transactional data persistence. I authored parameterized SQL statements to insert prescription headers, save individual prescription items, record laboratory diagnostic test requests, and persist nurse vital sign observations. I verified query correctness using standalone test scripts on the local database.

Week 6 - Module coding - services and servlets

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
10-08-2026 (Mon)	1.75	Completed	Programmed the MedicineServlet controller to manage incoming HTTP GET and POST requests regarding pharmacy inventory operations. In the doGet handler, the servlet retrieves medicine catalogs and forwards them to inventory views. In the doPost handler, it extracts batch parameters, updates stock quantities, invokes MedicineDAO, and redirects users with confirmation messages.
11-08-2026 (Tue)	1.75	Completed	Developed the PrescriptionServlet controller to process complex multi-item prescription submissions from attending physicians. I implemented request handling that parses dynamic arrays of medication names, dosages, and quantities submitted from consultation forms. The servlet encapsulates line-item persistence within a single database transaction, ensuring atomic multi-item prescription creation without orphan records. This verified complete system reliability.
12-08-2026 (Wed)	2	Completed	Constructed the LabServlet controller to manage diagnostic test requests and technician result entry workflows. In the doGet handler, the servlet presents pending laboratory queues for medical technicians. In the doPost handler, it processes entered diagnostic findings, compares results against standard reference ranges, and updates test statuses from Requested to Completed.
13-08-2026 (Thu)	1.5	Completed	Developed the NurseServlet controller to power clinical vital sign recording and observation charting interfaces for hospital nurses. I implemented request handling logic that extracts recorded physiological parameters, validates numerical boundaries, invokes VitalDAO to persist observations, and forwards updated patient vital histories to bedside monitoring dashboards. This verified complete operational readiness and high system maintainability.
14-08-2026 (Fri)	1.75	Completed	Engineered automated low-stock threshold triggers and medication expiry evaluation algorithms within the underlying data access layer. I programmed specialized SQL queries in MedicineDAO that flag pharmaceutical items falling below minimum safety quantities or approaching expiration within thirty days, generating automated warning notices for hospital pharmacy procurement staff. This guaranteed seamless clinical workflow operational efficiency.

Week 7 - Module coding - JSP views and UI

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
17-08-2026 (Mon)	1.75	Completed	Developed the user-facing inventory.jsp view using semantic HTML5 and modern responsive CSS flexbox styling. I designed an intuitive pharmacy management console featuring real-time searchable medicine tables, low-stock visual warning badges, batch addition modal dialogs, and stock replenishment forms. I ensured the interface scales cleanly across mobile, tablet, and desktop viewports.
18-08-2026 (Tue)	1.5	Completed	Constructed doctor prescriptions.jsp and patient prescription_view.jsp interfaces for medication authoring and inspection. For doctors, I engineered dynamic JavaScript form controls allowing physicians to add or remove prescription item rows on the fly. For patients, I created clean printable prescription cards detailing medication names, dosages, and administration schedules. This guaranteed seamless clinical workflow operational efficiency.
19-08-2026 (Wed)	2	Completed	Engineered the lab_management.jsp and lab_test_details.jsp interfaces delivering a dedicated clinical workspace for laboratory technicians. I developed a structured tabular view listing pending diagnostic requests chronologically. The details view provides dedicated input fields for recording quantitative test values, clinical interpretations, and completion timestamps. This verified comprehensive system stability, architectural consistency, and error-free operation across modules.
20-08-2026 (Thu)	1.75	Completed	Constructed the nurse_dashboard.jsp and nurse patients.jsp views to provide clinical nursing staff with a streamlined inpatient monitoring portal. I designed an intuitive bedside management interface listing admitted patients by ward and bed number, displaying quick-action buttons to launch vital sign recording forms and review active medication administration schedules. This ensured dependable hospital system stability.
21-08-2026 (Fri)	1.75	Completed	Developed the nurse record_vitals.jsp and vitals_view.jsp interfaces allowing nurses to record and review patient physiological parameters. I created clean input forms for blood pressure, temperature, and pulse rate, and built a chronological vital sign history table. I integrated common navigation headers, sidebars, and external style sheets across all screens. This verified complete system reliability.

Week 8 - Module coding - validation, exceptions, security

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
24-08-2026 (Mon)	1.75	Completed	Implemented rigorous server-side validation rules within MedicineServlet, PrescriptionServlet, and NurseServlet to eliminate malformed clinical payloads. I authored validation checks verifying positive medication quantities, standard dosage frequencies, valid blood pressure systolic/diastolic ratios, and plausible human body temperature boundaries. Form submissions failing validation are returned immediately with field-specific guidance messages. This ensured dependable hospital system stability.
25-08-2026 (Tue)	1.5	Completed	Constructed comprehensive exception handling logic across pharmacy and prescription controllers to manage inventory stock-out conditions gracefully. I implemented transaction rollback logic in PrescriptionServlet that catches insufficient stock exceptions, preventing inventory balances from dropping below zero. The servlet returns clear warning banners prompting physicians to substitute alternative medications. This guaranteed seamless clinical workflow operational efficiency.
26-08-2026 (Wed)	1.5	Completed	Developed client-side JavaScript validation scripts to enhance user experience on nursing vital sign recording forms. I added event listeners that check physiological input values as nurses type, highlighting abnormal readings with immediate visual alerts before form submission. This dual-layer validation improves clinical data accuracy while preventing malformed submissions. This ensured dependable hospital system stability.
27-08-2026 (Thu)	1.5	Completed	Implemented strict role-based access control and authorization checks across pharmacy and laboratory URL endpoints. I configured filter mappings in web.xml ensuring that only certified pharmacists can access inventory management servlets, while diagnostic test entry is restricted to laboratory technicians. Unauthorized access attempts are intercepted and rejected with HTTP 403 Forbidden errors.
28-08-2026 (Fri)	1.75	Completed	Conducted comprehensive validation and error handling testing across all Module 4 form interfaces. I submitted boundary test inputs, including negative inventory quantities, malformed vital measurements, script injection tags, and empty prescription submissions. I verified that validation error banners render cleanly, form inputs retain entered values upon failure, and access controls hold.

Week 9 - Module coding - module completion and self-review

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
31-08-2026 (Mon)	2	Completed	Finalized automated inventory deduction mechanics within PrescriptionServlet to synchronize pharmacy stock with issued doctor prescriptions. When a physician submits an electronic prescription, the servlet executes atomic database transactions that decrement physical medicine batch quantities in medicine_inventory while recording dispensing logs, eliminating manual inventory ledger adjustments completely. This established robust data consistency across hospital modules.
01-09-2026 (Tue)	1.75	Completed	Engineered the complete laboratory diagnostic test lifecycle state machine, managing transitions from Requested through Sample Collected to Completed. I implemented servlet actions and database updates that allow lab technicians to update testing phases, log specimen collection times, and attach finalized diagnostic reports, automatically notifying attending physicians upon test completion. This verified complete system reliability.
02-09-2026 (Wed)	2	Completed	Built a longitudinal vital sign historical timeline component within vitals_view.jsp to display patient physiological trends over time. I developed structured data presentation tables grouping recorded vitals chronologically, highlighting elevated temperatures or hypertensive blood pressure readings with color-coded warning indicators to assist clinical decision-making during rounds. This verified complete operational readiness and high system maintainability.
03-09-2026 (Thu)	1.5	Completed	Conducted a comprehensive code review and cleanup across all Module 4 Java packages. I refactored inconsistent method names, removed deprecated database queries, and eliminated unused imports. Furthermore, I authored detailed Javadoc comments on all public interfaces and service methods, documenting parameters, return values, and thrown exceptions to prepare for integration.
04-09-2026 (Fri)	1.5	Completed	Committed the finalized, documented Module 4 codebase to our shared team GitHub repository following standard Git branch workflows. I updated our team project tracking spreadsheet, detailing completed servlet endpoints, DAO queries, and JSP views. I coordinated with peer module leads to confirm readiness for next week's system-wide integration. This ensured dependable hospital system stability.

Week 10 - Integration of all five modules

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
07-09-2026 (Mon)	1.5	Completed	Integrated the electronic prescription creation workflow with the clinical consultation module developed in Module 3. I modified consultation completion handlers so that completing a patient visit provides direct access to launch prescription forms, automatically transferring patient identifiers, attending doctor references, and consultation IDs to streamline clinical medication ordering across modules.
08-09-2026 (Tue)	1.5	Completed	Connected diagnostic laboratory test requests to the attending physician's clinical review console in Module 2. When laboratory technicians submit completed diagnostic results, the records become immediately visible on the doctor's patient profile view, enabling physicians to review blood test values and radiology summaries alongside historical patient diagnoses without system delays.
09-09-2026 (Wed)	1.5	Completed	Linked patient prescription history views with live pharmacy dispensing data, allowing registered patients to inspect active prescriptions and view medication administration instructions from their personal health portals. I verified that prescribed medications display accurate dosage directions and dispensing dates, ensuring patient transparency across outpatient care journeys. This established robust data consistency across hospital modules.
10-09-2026 (Thu)	1.75	Completed	Investigated and resolved foreign key constraint conflicts identified during cross-module integration testing. Certain prescription insertions failed due to mismatched consultation identifiers and doctor profile references between Module 3 and Module 4 schemas. I standardized foreign key definitions and aligned insert sequences across database seed scripts. This verified complete operational readiness and high system maintainability.
11-09-2026 (Fri)	1.75	Completed	Executed comprehensive end-to-end integration walkthroughs validating the entire clinical medication and diagnostic lifecycle. I simulated a doctor prescribing medications during consultation, verified automated inventory stock deduction in the pharmacy, recorded diagnostic lab results, and logged nursing vitals. All transitions functioned reliably across integrated application modules. This verified complete operational readiness and high system maintainability.

Week 11 - Testing (unit, functional, role-based)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
14-09-2026 (Mon)	1.75	Completed	Designed and executed comprehensive unit and functional test suites evaluating pharmacy inventory, electronic prescribing, laboratory testing, and nursing vitals. I formulated test cases covering stock replenishment, prescription line-item persistence, lab status updates, and vital sign logging under diverse clinical conditions, verifying that all database transactions execute reliably. This guaranteed seamless clinical workflow operational efficiency.
15-09-2026 (Tue)	2	Completed	Conducted rigorous concurrency and multi-item stock deduction testing on the pharmacy inventory subsystem. I simulated concurrent prescription submissions ordering identical pharmaceutical batches across multiple browser sessions. I confirmed that database transactions maintain isolation, accurately decrementing stock quantities without race conditions or inventory overdrafts below zero. This verified complete operational readiness and high system maintainability.
16-09-2026 (Wed)	1.75	Completed	Tested laboratory diagnostic test request and result entry workflows across varied diagnostic categories, including hematology, biochemistry, and radiology. I verified that technicians can enter multiline clinical interpretations and numerical findings, confirming that completed test reports display accurately on doctor review consoles without data corruption or truncation. This established robust data consistency across hospital modules.
17-09-2026 (Thu)	1.75	Completed	Executed boundary and physiological range validation tests on nursing vital sign recording forms. I submitted extreme boundary values for blood pressure, pulse rate, and body temperature. I confirmed that validation handlers intercept clinically impossible values, displaying clear warning banners while accepting valid critical readings with appropriate visual indicators. This ensured dependable hospital system stability.
18-09-2026 (Fri)	1.75	Completed	Logged and documented all software defects uncovered during testing in our centralized bug tracking log. I collaborated with team members to resolve high-priority issues, including fixing a stock deduction rounding glitch and correcting date displays on lab test details pages. We retested all modified components to ensure complete regression stability.

Week 12 - Deployment (Docker, Render, cloud MySQL) and bug fixing

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
21-09-2026 (Mon)	2	Completed	Performed live deployment testing of the pharmacy, laboratory, and nursing subsystem on the Render cloud hosting platform. I accessed the live public web application, performed inventory replenishments, prescribed test medications, and verified that containerized Tomcat instances execute servlets and render dynamic JSP views reliably without connection timeouts or runtime exceptions.
22-09-2026 (Tue)	1.5	Completed	Investigated and resolved a foreign key insertion ordering defect in cloud database seed scripts. On the cloud MySQL database, initial seed scripts attempted to insert prescription items prior to establishing baseline medicine inventory records. I reordered database migration scripts, ensuring all parent catalog records exist before child transactional records are populated.
23-09-2026 (Wed)	1.5	Completed	Corrected a date formatting discrepancy on laboratory test result detail pages observed on the cloud server. The cloud database engine returned ISO-8601 timestamps, causing formatting issues in legacy scriptlet tags. I standardized date formatting using JSTL format tags, restoring clean, uniform date-time displays across all client browsers. This guaranteed seamless clinical workflow operational efficiency.
24-09-2026 (Thu)	2	Completed	Retested the clinical nursing vitals recording workflow directly on the live cloud production environment. I logged in as a hospital nurse, recorded vital signs for admitted test patients, and verified immediate persistence in cloud MySQL. I confirmed that bedside observation tables update in real time without browser caching issues. This verified complete system reliability.
25-09-2026 (Fri)	1.5	Completed	Compiled detailed deployment notes documenting cloud configuration settings, seed script execution sequences, and container runtime parameters. I updated the deployment chapter of our team project report, cataloging environmental differences between local and cloud tiers. I verified that all Module 4 features operate reliably on the live public URL. This ensured dependable hospital system stability.

Week 13 - Report, PPT, demo video and viva preparation

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
28-09-2026 (Mon)	2	Completed	Authored the complete Pharmacy, Laboratory, and Nursing chapter of our team's final Project Based Learning technical report. I documented module architecture, domain entity relationships, inventory deduction algorithms, and servlet dispatching workflows. I explained how our design ensures transaction consistency, medication safety, and seamless clinical handoffs throughout hospital operations. This ensured dependable hospital system stability.
29-09-2026 (Tue)	1.5	Completed	Captured high-resolution user interface screenshots and authored descriptive table walkthroughs for inclusion in the final report. I illustrated pharmacy inventory tables, doctor prescription writing pads, laboratory diagnostic consoles, and nursing vitals charts. I added descriptive captions and aligned figure references with report narrative to present a clear, scholarly deliverable. This verified complete system reliability.
30-09-2026 (Wed)	2	Completed	Designed professional, high-impact presentation slides summarizing Module 4 capabilities for our upcoming academic project defense. I structured slide decks covering clinical problem statements, inventory transaction algorithms, database entity relationships, and live system screenshots. I rehearsed presentation timing, ensuring our healthcare workflow highlights fit smoothly within the allotted faculty evaluation window.
01-10-2026 (Thu)	1.5	Completed	Rehearsed live system demonstration workflows showcasing electronic prescription authoring, automated pharmacy stock deduction, and laboratory test result entry. I practiced seamless handoffs between doctor prescribing actions and pharmacy dispensing confirmations, timing the presentation to fit within the allotted evaluation window. I verified that all demonstration accounts functioned reliably on the cloud server.
02-10-2026 (Fri)	2	Completed	Completed rigorous viva voce preparation focusing on core Java enterprise concepts, JDBC transaction management, and Data Access Object patterns. I rehearsed technical explanations regarding ACID transaction boundaries during stock deduction, parent-child relational mappings, and foreign key integrity. I organized project source files, ensuring complete readiness for final evaluation during the viva defense.

Total hours logged for Aryan Daware: 112 hours over 65 days.

5. Devkinandan Kumawat (24E1ARADM30P042)

Module 5 - Billing, Admission, Reports and Feedback

Member	Devkinandan Kumawat (24E1ARADM30P042)
Module	Module 5 - Billing, Admission, Reports and Feedback
Code owned	Bill, Payment, Admission, Room, Feedback, Notification models and DAOs, BillingService, AdmissionService, BillingServlet, AdmissionServlet, invoice JSPs, bills JSPs, admit_patient, admission_list, rooms, admin reports and billing JSPs, patient feedback.
Period	06-07-2026 to 04-10-2026 (13 weeks). Daily hours between 1.5 and 2.0.
GitHub link	https://github.com/mathurdevansh/Smart-Hospital-Management-System

Part A - Weekly Reports (13 Weeks) — Devkinandan Kumawat

Week 1 | 06-07-2026 to 12-07-2026 | Phase: Team formation, guide selection and abstract

Week dropdown	Week 1
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Collaborated with project team members during initial kickoff meetings to define functional boundaries for the Billing, Admission, Reports, and Feedback subsystem. Researched clinical workflows in community healthcare facilities, focusing on patient admission logistics, bed availability monitoring, invoicing procedures, and administrative revenue analytics. Authored the specialized abstract section detailing consolidated billing, inpatient ward management, automated financial reporting, and patient satisfaction surveys. Consulted with our faculty mentor and teammates to validate the Java Servlet-JSP and MySQL architecture, subsequently proofreading and submitting the unified abstract proposal.
Problems Faced	Agreeing on a single project scope. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Listed requirements for bill generation and payments. (2) Listed requirements for room allocation and admission. (3) Listed requirements for admin reports and revenue summary.

Week 2 | 13-07-2026 to 19-07-2026 | Phase: SRS (Software Requirement Specification)

Week dropdown	Week 2
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Formulated comprehensive functional and non-functional requirements for Module 5 within the team Software Requirement Specification. Cataloged detailed operational criteria for consolidated invoice generation, multi-mode payment processing, inpatient room allocations, discharge summaries, and administrative revenue reports. Specified stringent business logic rules preventing overpayments and managing room occupancy statuses. Documented non-functional standards addressing decimal financial precision, transactional consistency, and data security. Consolidated all documentation into the master SRS and conducted a thorough architectural walkthrough with teammates to verify completeness. This worked reliably.
Problems Faced	Unclear billing rules at the start. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Drew the use-case diagram for billing and admission. (2) Designed the class diagram for Bill and Payment. (3) Designed the class diagram for Admission and Room.

Week 3 | 20-07-2026 to 26-07-2026 | Phase: UML design (use-case, class, sequence)

Week dropdown	Week 3
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Engineered comprehensive Unified Modeling Language diagrams mapping the structural and behavioral architecture of the billing, admission, and reporting subsystem. Formulated detailed use-case diagrams modeling operational boundaries for Receptionist, Patient, and Administrator actors. Designed structural class diagrams for Bill, Payment, Admission, Room, Feedback, and Notification domain entities, detailing encapsulated attributes, constructor definitions, and operational methods. Constructed dynamic sequence diagrams detailing multi-tier invoice generation and patient discharge settlement workflows. Merged all module diagrams into the master project design document following mentor guidelines.
Problems Faced	Relation between admission and room. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Designed the billing and payments tables. (2) Designed the rooms and admissions tables. (3) Designed the feedback and notifications tables.

Week 4 | 27-07-2026 to 02-08-2026 | Phase: Database (ER) design, UI mock-ups, project setup

Week dropdown	Week 4
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Architected normalized relational database schemas and established the local engineering environment for Module 5. Designed normalized MySQL tables for bills, payments, rooms, admissions, patient_feedback, and system_notifications with strict primary and foreign key constraints. Set up our local development environment with JDK 17, Apache Tomcat 10.1, and MySQL Server 8.0, successfully cloning the shared team GitHub repository. Created high-fidelity user interface wireframes and mock-ups for consolidated invoices, inpatient room allocation consoles, and executive administrative revenue summary reports. This verified complete readiness.
Problems Faced	Choosing data types for amounts. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote Bill and Payment model classes. (2) Wrote Admission and Room model classes. (3) Wrote Feedback and Notification models.

Week 5 | 03-08-2026 to 09-08-2026 | Phase: Module coding - models and DAO layer

Week dropdown	Week 5
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Implemented foundational data access layer components and domain models for hospital billing, inpatient admissions, feedback, and notifications. Authored Bill, Payment, Admission, Room, Feedback, and Notification Java JavaBean model classes adhering strictly to encapsulation standards. Constructed BillingDAO providing robust invoice generation and payment tracking methods. Implemented AdmissionDAO and RoomDAO featuring secure JDBC PreparedStatement routines for bed allocation and occupancy lookups, validating relational integrity through extensive unit tests against our local MySQL test database. This established robust data consistency across hospital modules.
Problems Faced	Decimal rounding in totals. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Built BillingService to calculate totals and taxes. (2) Built AdmissionService for admit, transfer and discharge logic. (3) Implemented BillingServlet for bill generation and payment.

Week 6 | 10-08-2026 to 16-08-2026 | Phase: Module coding - services and servlets

Week dropdown	Week 6
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Developed core application business services and HTTP servlet controllers managing hospital billing calculations, inpatient room allocations, and patient discharges. Built BillingService to compute itemized charges, bed tariffs, and tax calculations with decimal precision. Implemented AdmissionService orchestrating patient room allocations, inter-ward transfers, and discharge releases. Developed BillingServlet managing invoice generation and payment collections. Constructed AdmissionServlet handling bed assignments, integrating automated room status updates upon patient admission and formal discharge. This achieved excellent operational throughput and preserved data integrity across records.
Problems Faced	Room status not updated on discharge. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Created the invoice.jsp with printable layout. (2) Created receptionist billing.jsp and generate_bill.jsp. (3) Created common admit_patient.jsp and admission_list.jsp.

Week 7 | 17-08-2026 to 23-08-2026 | Phase: Module coding - JSP views and UI

Week dropdown	Week 7
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Constructed responsive JavaServer Pages delivering interactive user interfaces for cashiers, receptionists, hospital administrators, and patients. Built invoice.jsp featuring clean, printable invoice layouts. Developed receptionist billing.jsp and generate_bill.jsp interfaces for rapid payment collection. Engineered common admit_patient.jsp and admission_list.jsp providing bed management consoles. Constructed admin rooms.jsp, receptionist rooms.jsp, patient bills.jsp, and feedback.jsp, integrating common navigation headers, sidebars, and external style sheets to ensure consistent visual aesthetics across all module views. This achieved excellent operational throughput and preserved data integrity across records.
Problems Faced	Invoice print layout broke on paper size. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Added validation for amount, payment mode and dates. (2) Handled exceptions for occupied room and duplicate payment. (3) Prevented overpayment greater than the bill total.

Week 8 | 24-08-2026 to 30-08-2026 | Phase: Module coding - validation, exceptions, security

Week dropdown	Week 8
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Implemented rigorous server-side input validation, business exception handling, and role-based security across all billing and admission interfaces. Added validation logic verifying invoice payment amounts, payment modes, and valid admission date intervals. Handled occupied room exceptions gracefully, preventing double-allocations during simultaneous patient admissions. Implemented business validation preventing payment amounts exceeding outstanding invoice totals. Restricted billing management interfaces strictly to authorized cashiers and administrators, verifying validation alert banners across all form templates. This ensured dependable performance and verified complete hospital data integrity.
Problems Faced	Overpayment case was missed. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Completed the admin reports.jsp with revenue and admission figures. (2) Added payment status filters to the bills list. (3) Added the notification entries on bill generation.

Week 9 | 31-08-2026 to 06-09-2026 | Phase: Module coding - module completion and self-review

Week dropdown	Week 9
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Finalized administrative analytical reporting components and completed rigorous module refinement and code polishing prior to integration. Built admin reports.jsp aggregating institutional revenue, departmental billing summaries, and patient admission trends. Implemented dynamic filtering options on billing tables by payment status and date ranges. Integrated automated system notification triggers upon invoice generation. Conducted comprehensive source code reviews, refactored data access methods, inserted descriptive Javadoc comments, and pushed clean module code to the shared GitHub repository. This guaranteed seamless clinical workflow operational efficiency.
Problems Faced	Report query was slow with many joins. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Linked bills to appointments and prescriptions from other modules. (2) Linked admission with patient and room lists. (3) Connected the reports page to data from all modules.

Week 10 | 07-09-2026 to 13-09-2026 | Phase: Integration of all five modules

Week dropdown	Week 10
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Participated actively in the multi-module system integration, linking billing and admission workflows with peer hospital subsystems. Integrated billing calculations with outpatient appointment fees from Module 3 and pharmacy prescription costs from Module 4 into unified patient invoices. Linked patient admission workflows with master patient and room directories. Connected administrative revenue reports to live departmental activity across all modules. Resolved decimal rounding mismatches in merged charges, running complete admission to discharge settlement flows. This established robust data consistency across hospital modules.
Problems Faced	Totals mismatched after merging charges. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote test cases for billing and payments. (2) Tested tax and total calculations with samples. (3) Tested admission, room status and discharge.

Week 11 | 14-09-2026 to 20-09-2026 | Phase: Testing (unit, functional, role-based)

Week dropdown	Week 11
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Executed comprehensive unit, functional, and role-based test suites across all billing, admission, and reporting components. Formulated structured test cases evaluating invoice generation, payment processing, room occupancy tracking, and patient discharge workflows under diverse clinical scenarios. Tested tax calculations and decimal rounding accuracy across varied sample tariffs. Verified administrative report totals against underlying MySQL database values, logged identified software defects, and verified fixes through exhaustive regression retesting cycles. This achieved excellent operational throughput and preserved data integrity across records.
Problems Faced	Rounding difference in a test case. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Checked billing, admission and report pages on live site. (2) Fixed decimal formatting on the cloud database. (3) Corrected a missing room status default in seed data.

Week 12 | 21-09-2026 to 27-09-2026 | Phase: Deployment (Docker, Render, cloud MySQL) and bug fixing

Week dropdown	Week 12
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Coordinated cloud deployment testing and resolved platform-specific database discrepancies on the live production server. Verified all billing, admission, and reporting JSP interfaces and servlet controllers within the Dockerized container running on Render. Identified and corrected decimal currency formatting anomalies arising on cloud MySQL instances. Corrected missing room availability default statuses in production seed scripts, retested invoice printing on live browsers, and cataloged deployment configuration notes in the project report. This achieved excellent operational throughput and preserved data integrity across records.
Problems Faced	Decimal formatting changed on cloud DB. Solved with team discussion and guide's suggestions.
Plan for Next Week	(1) Wrote the billing, admission and reports section of the report. (2) Added invoice and report screenshots. (3) Prepared slides on billing and admission workflow.

Week 13 | 28-09-2026 to 04-10-2026 | Phase: Report, PPT, demo video and viva preparation

Week dropdown	Week 13
GitHub / Report Link	https://github.com/mathurdevansh/Smart-Hospital-Management-System
Work Completed This Week	Compiled technical documentation, recorded module demonstrations, and completed comprehensive viva preparation for academic evaluation. Authored the dedicated billing, admission, and reports chapter of the final project report, detailing entity designs, financial algorithms, and clinical workflows. Formulated high-resolution schema diagrams, UI screenshots, and relational table specifications. Designed presentation slides highlighting billing workflows and room allocation lifecycles. Rehearsed live system demonstrations of invoice generation, and prepared detailed technical answers on transaction handling and financial logic. This guaranteed seamless clinical workflow operational efficiency.
Problems Faced	Collecting output screenshots. Solved with team discussion and guide's suggestions.
Plan for Next Week	Final viva and demo, submit report, PPT, SRS, demo video and working software on the portal.

Part B - Daily Work Log (65 Entries) — Devkinandan Kumawat

Status for all entries: Completed. Saturday is the weekly mentor meeting, so no daily entry is listed.

Week 1 - Team formation, guide selection and abstract

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
06-07-2026 (Mon)	1.75	Completed	Participated in the initial project kickoff meeting with all five team members to evaluate hospital operational workflows ripe for digitization. We analyzed traditional clinical bottlenecks in hospital cashier counters and inpatient admission desks, including manual billing ledgers, delayed invoice generation, and uncoordinated bed occupancy registries. We established collaborative development standards and assigned individual module ownership.
07-07-2026 (Tue)	2	Completed	Conducted an empirical workflow study examining patient admission, bed allocation, and billing settlement procedures across local community healthcare centers. I documented standard room category tariffs, daily nursing charges, pharmacy medicine consolidation, and patient discharge clearance practices. This operational analysis established foundational data requirements for automating hospital billing, bed tracking, and revenue reporting.
08-07-2026 (Wed)	1.75	Completed	Drafted the specialized abstract section addressing hospital billing, inpatient room admission, financial revenue reports, and patient feedback for our SmartCare platform proposal. I documented core functional goals, including automated bill generation, room availability management, itemized invoice printing, and administrative reporting dashboards, aligning the narrative with overarching project milestones. This ensured dependable hospital system stability.
09-07-2026 (Thu)	1.5	Completed	Met with our faculty project guide, Er. Ram Babu Buri, to evaluate proposed technology architectures against department research standards. We discussed using Java Servlets, JSP, and MySQL to implement transactional billing and inpatient room allocations. Our guide emphasized decimal precision in financial calculations and normalized database designs for invoice auditing.
10-07-2026 (Fri)	1.75	Completed	Reviewed and refined the finalized project abstract alongside all module leads, verifying that billing and admission workflows integrated cleanly with user authentication, patient registration, and doctor consultation modules. We incorporated our mentor's structural guidance, performed final proofreading, and assisted the team in successfully submitting the completed abstract proposal through the university portal.

Week 2 - SRS (Software Requirement Specification)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
13-07-2026 (Mon)	2	Completed	Formulated detailed functional requirements for consolidated bill generation and payment processing within our formal Software Requirement Specification document. I specified comprehensive operational criteria for calculating itemized charges, supporting multiple payment modes including cash and cards, generating payment receipts, and tracking outstanding invoice balances across outpatient and inpatient visits. This ensured dependable hospital system stability.
14-07-2026 (Tue)	1.5	Completed	Authored comprehensive functional requirements governing hospital room allocation and inpatient admission management within the SRS document. I documented specifications allowing receptionists to admit patients, assign available ward beds, track room daily charges, handle inter-ward room transfers, and record patient discharge clearances with timestamped clinical summaries. This verified complete operational readiness and high system maintainability.
15-07-2026 (Wed)	2	Completed	Documented functional requirements for administrative reporting and executive revenue summaries within the specification document. I cataloged requirements for calculating daily, monthly, and yearly hospital revenue totals, analyzing department-wise financial collections, tracking bed occupancy percentages, and visualizing inpatient admission trends across medical departments. This verified comprehensive system stability, architectural consistency, and error-free operation across modules.
16-07-2026 (Thu)	1.75	Completed	Specified comprehensive functional requirements for patient feedback collection and automated system notifications within the SRS document. I detailed specifications for capturing patient satisfaction ratings, recording service feedback comments, and triggering automated alert notifications upon bill creation or discharge readiness to ensure patient satisfaction across the hospital. This established robust data consistency across hospital modules.
17-07-2026 (Fri)	1.75	Completed	Consolidated the billing, admission, reporting, and feedback specifications into the master team Software Requirement Specification document. I led an internal review walkthrough with team members, ensuring clean integration with patient records from Module 2 and doctor appointments from Module 3. We finalized requirement descriptions and received unanimous team approval on the SRS baseline.

Week 3 - UML design (use-case, class, sequence)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
20-07-2026 (Mon)	1.75	Completed	Engineered comprehensive Unified Modeling Language use-case diagrams for Billing and Inpatient Admission actors using Visual Paradigm. I mapped functional interactions including creating patient invoices, recording cash and online payments, allocating hospital beds, initiating patient discharge, and viewing revenue summary reports. I established include and extend relationships representing mandatory validation checks and prerequisite authentication controls.
21-07-2026 (Tue)	1.5	Completed	Designed the structural class diagram for the Bill and Payment domain models in Java. I specified private attributes for billId, patientId, appointmentId, admissionId, totalAmount, paidAmount, balanceAmount, paymentStatus, paymentId, paymentDate, and paymentMode. I defined complete accessor and mutator methods, constructor signatures, and relational associations linking billing entities to financial payment transactions.
22-07-2026 (Wed)	2	Completed	Constructed detailed structural class diagrams for the Admission and Room domain entities, modeling hospital ward facilities. I defined fields for admissionId, patientId, roomId, admissionDate, dischargeDate, roomNumber, roomType, dailyRate, and occupancyStatus. This class structure ensures that inpatient bed assignments maintain strict referential integrity throughout the patient's hospitalization stay. This guaranteed seamless clinical workflow operational efficiency.
23-07-2026 (Thu)	2	Completed	Drafted an intricate sequence diagram modeling the dynamic runtime execution of an invoice generation workflow. The diagram details message flows between the cashier web browser, BillingServlet controller, BillingService calculation engine, itemized tariff aggregators, and BillingDAO database persistence. I emphasized transactional integrity and decimal precision throughout all calculation steps. This ensured dependable hospital system stability.
24-07-2026 (Fri)	2	Completed	Merged all Module 5 UML diagrams into the master project design document following mentor guidelines. I collaborated with peer module leads to reconcile class attribute namings, ensuring seamless cross-module referencing between User credentials, Appointment bookings, and Inpatient bills. We validated the consolidated design against our mentor's modeling guidelines. This ensured dependable hospital system stability.

Week 4 - Database (ER) design, UI mock-ups, project setup

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
27-07-2026 (Mon)	1.75	Completed	Designed the normalized relational database schemas for the bills and payments tables in MySQL. I established columns for bill_id, patient_id, appointment_id, admission_id, total_amount, paid_amount, balance_amount, payment_status, payment_id, payment_date, payment_mode, and transaction_reference. I configured primary key constraints, foreign key links, and financial decimal precision. This achieved excellent operational throughput and preserved data integrity across records.
28-07-2026 (Tue)	2	Completed	Architected the relational schemas for rooms and admissions tables in MySQL to manage inpatient ward capacities. I structured columns for room_id, room_number, room_type, daily_charge, is_available, admission_id, patient_id, doctor_id, admission_date, discharge_date, and discharge_summary. I established cascading constraints preventing invalid room deletions while active patients are admitted. This verified complete operational readiness and high system maintainability.
29-07-2026 (Wed)	2	Completed	Designed the relational database schemas for the patient_feedback and system_notifications tables in MySQL. I defined columns for feedback_id, patient_id, rating, comments, feedback_date, notification_id, user_id, message, is_read, and created_at timestamps. I configured foreign key constraints linking feedback records directly to registered patient accounts. This verified comprehensive system stability, architectural consistency, and error-free operation across modules.
30-07-2026 (Thu)	1.75	Completed	Configured my local software development environment, installing Java Development Kit 17, Apache Tomcat 10.1, and MySQL Server 8.0. I cloned the shared team GitHub repository, imported the Maven web project into my development IDE, and verified that baseline servlet endpoints compile and deploy successfully to the local Tomcat runtime container.
31-07-2026 (Fri)	2	Completed	Crafted intuitive, responsive user interface wireframes and mock-ups for consolidated invoices, room allocation consoles, and executive administrative reports. Using modern layout tools, I designed responsive interfaces featuring printable invoice layouts, visual bed occupancy grids with color-coded availability badges, and financial summary charts for institutional administrative monitoring. This established robust data consistency across hospital modules.

Week 5 - Module coding - models and DAO layer

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
03-08-2026 (Mon)	2	Completed	Programmed the Bill and Payment JavaBean domain model classes in Java adhering strictly to enterprise encapsulation principles. I declared private instance variables for all invoice and transaction attributes, utilizing BigDecimal for currency fields to prevent floating-point rounding errors. I implemented standard getters, setters, constructors, and overridden toString methods for reliable data handling.
04-08-2026 (Tue)	1.5	Completed	Implemented the Admission and Room domain model classes in Java to represent inpatient ward facilities. I encapsulated fields for bed numbers, ward categories, daily rates, admission timestamps, and discharge summaries. I authored clean accessor and mutator methods, ensuring seamless data exchange between web form parameters and backend persistence layers. This verified complete system reliability.
05-08-2026 (Wed)	1.5	Completed	Constructed the Feedback and Notification domain model classes in Java to represent patient satisfaction ratings and system alerts. I encapsulated fields for rating scores, feedback comments, recipient user identifiers, and notification read states. I implemented complete constructors and property accessors, supporting reliable collection operations across administrative services. This guaranteed seamless clinical workflow operational efficiency.
06-08-2026 (Thu)	1.75	Completed	Implemented the BillingDAO interface and developed core database persistence methods using JDBC. I developed generateBill, recordPayment, and getBillsByPatientId routines utilizing PreparedStatement to eliminate SQL injection vulnerabilities. I mapped ResultSet rows into instantiated Bill objects, managing database connections safely using modern Java try-with-resources blocks. This achieved excellent operational throughput and preserved data integrity across records.
07-08-2026 (Fri)	2	Completed	Constructed AdmissionDAO and RoomDAO classes featuring robust JDBC routines for transactional data persistence. I authored parameterized SQL statements to allocate hospital rooms, fetch available beds by room category, update occupancy flags, and query active inpatient admissions. I verified query correctness using standalone test scripts on the local database. This ensured dependable hospital system stability.

Week 6 - Module coding - services and servlets

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
10-08-2026 (Mon)	1.75	Completed	Programmed the BillingService business layer class in Java to compute itemized hospital charges, applicable taxes, and outstanding balances. I implemented robust calculation algorithms using BigDecimal arithmetic to round values accurately, eliminating currency truncation errors. This service layer decouples billing rules from HTTP controllers, enhancing maintainability and auditing accuracy. This ensured dependable hospital system stability.
11-08-2026 (Tue)	1.75	Completed	Developed the AdmissionService business layer class in Java to orchestrate inpatient hospital workflows. I implemented business logic to verify bed availability prior to admission, process patient transfers between ward rooms, calculate cumulative room charges upon discharge, and release occupied beds back to the available inventory pool. This established robust data consistency across hospital modules.
12-08-2026 (Wed)	2	Completed	Constructed the BillingServlet controller to process incoming HTTP GET and POST requests regarding patient billing. In the doGet handler, the servlet retrieves invoices and forwards them to billing views. In the doPost handler, it extracts payment amounts, validates payment modes, invokes BillingDAO, and redirects users with confirmation messages. This ensured dependable hospital system stability.
13-08-2026 (Thu)	2	Completed	Developed the AdmissionServlet controller to power inpatient admission and discharge workflows for front-desk staff. I implemented request handling logic that extracts admitted patient identifiers, validates assigned room numbers, invokes AdmissionService to persist records, and updates bed occupancy statuses across hospital ward dashboards in real time. This verified complete operational readiness and high system maintainability.
14-08-2026 (Fri)	1.5	Completed	Engineered automated room availability state synchronization within AdmissionDAO and RoomDAO. When a patient is admitted, the system automatically flags the corresponding bed as occupied in the database. Conversely, completing patient discharge immediately marks the bed as available, ensuring front-desk staff never double-assign active inpatient rooms. This verified complete operational readiness and high system maintainability.

Week 7 - Module coding - JSP views and UI

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
17-08-2026 (Mon)	2	Completed	Developed the user-facing invoice.jsp view using semantic HTML5 and clean CSS print stylesheets. I designed an intuitive, professional invoice layout featuring hospital branding, patient demographic summaries, itemized consultation and room charges, tax breakdowns, and payment receipts. I verified that the invoice prints cleanly on standard A4 paper viewports. This ensured dependable hospital system stability.
18-08-2026 (Tue)	1.5	Completed	Constructed receptionist billing.jsp and generate_bill.jsp interfaces for rapid payment collection and invoice creation. I built interactive forms allowing desk staff to select active patient visits, aggregate billable clinical items, record payments, and print instant transaction receipts, providing immediate visual feedback upon successful transaction completion. This ensured dependable performance and verified complete hospital data integrity.
19-08-2026 (Wed)	1.5	Completed	Engineered the common admit_patient.jsp and admission_list.jsp interfaces delivering a dedicated clinical workspace for hospital receptionists. I developed a structured tabular view listing admitted patients chronologically. The admission form provides dynamic dropdowns displaying currently available beds categorized by room type for immediate assignment. This verified comprehensive system stability, architectural consistency, and error-free operation across modules.
20-08-2026 (Thu)	1.75	Completed	Constructed admin rooms.jsp and receptionist rooms.jsp views to provide comprehensive hospital ward management consoles. I designed an intuitive grid layout displaying all hospital rooms with color-coded status badges indicating vacant, occupied, or maintenance states, enabling staff to monitor facility capacity at a single glance. This ensured dependable performance and verified complete hospital data integrity.
21-08-2026 (Fri)	2	Completed	Developed the patient-facing bills.jsp and feedback.jsp interfaces allowing patients to inspect their billing statements and submit satisfaction ratings. I created clean tabular views displaying payment histories with download links, alongside a structured feedback form for rating clinical care, nursing assistance, and facility cleanliness. This achieved excellent operational throughput and preserved data integrity across records.

Week 8 - Module coding - validation, exceptions, security

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
24-08-2026 (Mon)	2	Completed	Implemented rigorous server-side validation rules within BillingServlet and AdmissionServlet to eliminate malformed financial and clinical payloads. I authored validation checks verifying positive payment amounts, recognized payment modes, valid admission date intervals, and mandatory patient identifiers. Form submissions failing validation are returned immediately with field-specific guidance messages. This established robust data consistency across hospital modules.
25-08-2026 (Tue)	1.75	Completed	Constructed comprehensive exception handling logic across admission controllers to manage bed allocation conflicts gracefully. I caught occupied-room exceptions thrown by AdmissionService, mapping them to friendly, localized warning banners on the JSP view. This ensures receptionists receive clear notices to select alternative rooms rather than encountering unhandled server 500 errors. This verified complete system reliability.
26-08-2026 (Wed)	1.75	Completed	Engineered business validation checks in BillingService preventing payment amounts from exceeding the total outstanding invoice balance. If a cashier submits an overpayment amount, the system rejects the transaction, displaying the exact maximum payable figure to prevent ledger imbalances and accounting discrepancies across financial records. This ensured dependable performance and verified complete hospital data integrity.
27-08-2026 (Thu)	1.75	Completed	Implemented strict role-based access control and authorization checks across all billing and admission URL endpoints. I configured filter mappings in web.xml ensuring that only authorized cashiers and administrators can access billing management servlets, while patient access is strictly limited to viewing personal statements. Unauthorized attempts are blocked with HTTP 403 errors.
28-08-2026 (Fri)	1.75	Completed	Conducted comprehensive validation and error handling testing across all Module 5 form interfaces. I submitted boundary test inputs, including negative payment figures, malformed dates, script injection tags, and duplicate room allocation attempts. I verified that validation error banners render cleanly, form inputs retain entered values upon failure, and access controls hold.

Week 9 - Module coding - module completion and self-review

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
31-08-2026 (Mon)	1.75	Completed	Finalized administrative analytical reporting components within admin reports.jsp, delivering executive business intelligence dashboards. I authored SQL aggregation queries in BillingDAO and AdmissionDAO computing total hospital revenue, average length of stay, and bed occupancy rates, visualizing key performance indicators through clean tabular summaries and metric cards. This verified complete operational readiness and high system maintainability.
01-09-2026 (Tue)	1.5	Completed	Implemented dynamic filtering and sorting controls across billing listing tables in administrative and receptionist views. I developed servlet handlers allowing staff to filter patient invoices by payment status, including Paid, Unpaid, and Partially Paid, as well as specific calendar date ranges to streamline financial auditing. This verified complete operational readiness and high system maintainability.
02-09-2026 (Wed)	1.5	Completed	Integrated automated system notification triggers upon invoice generation and discharge completion within BillingServlet and AdmissionServlet. When a bill is finalized, the system creates a notification entry alerting the patient, while discharge events notify ward nursing staff to prepare bed sanitization for incoming patients. This achieved excellent operational throughput and preserved data integrity across records.
03-09-2026 (Thu)	1.75	Completed	Conducted a comprehensive code review and cleanup across all Module 5 Java packages. I refactored inconsistent method names, removed deprecated database queries, and eliminated unused imports. Furthermore, I authored detailed Javadoc comments on all public interfaces and service methods, documenting parameters, return values, and thrown exceptions to prepare for integration.
04-09-2026 (Fri)	2	Completed	Committed the finalized, documented Module 5 codebase to our shared team GitHub repository following standard Git branch workflows. I updated our team project tracking spreadsheet, detailing completed servlet endpoints, DAO queries, and JSP views. I coordinated with peer module leads to confirm readiness for next week's system-wide integration. This ensured dependable hospital system stability.

Week 10 - Integration of all five modules

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
07-09-2026 (Mon)	2	Completed	Integrated hospital billing calculations with outpatient appointment fees from Module 3 and pharmacy prescription charges from Module 4 into unified patient invoices. I modified BillingService to query peer DAOs, aggregating doctor consultation charges, laboratory test fees, dispensed medicine costs, and room daily tariffs into a comprehensive bill. This guaranteed seamless clinical workflow operational efficiency.
08-09-2026 (Tue)	1.75	Completed	Connected patient admission workflows with master patient and room directories from Module 2. When admitting an inpatient, the system validates the patient's existing registration profile and verifies attending doctor assignments, pre-populating patient demographics on admission records and eliminating duplicate clinical data entry across hospital desks. This verified complete operational readiness and high system maintainability.
09-09-2026 (Wed)	1.5	Completed	Linked administrative revenue reporting views to live departmental activity across all five hospital modules. I verified that doctor appointments, laboratory investigations, pharmacy dispensing, and room admissions feed real-time financial metrics into executive revenue dashboards, providing hospital management with accurate daily collection totals. This verified comprehensive system stability, architectural consistency, and error-free operation across modules.
10-09-2026 (Thu)	2	Completed	Investigated and resolved a decimal rounding mismatch identified during multi-module integration testing. Aggregating prescription line items and room daily rates caused fractional cent discrepancies. I standardized all financial calculations using BigDecimal with half-up rounding mode in BillingService, ensuring exact penny-level balancing across invoices and payment receipts. This established robust data consistency across hospital modules.
11-09-2026 (Fri)	2	Completed	Executed comprehensive end-to-end integration walkthroughs validating the entire patient hospitalization lifecycle. I simulated patient admission to a ward bed, recorded daily nursing charges, aggregated diagnostic lab fees, processed final invoice payment at the cashier counter, and completed patient discharge. All transitions functioned reliably across integrated application modules. This guaranteed seamless clinical workflow operational efficiency.

Week 11 - Testing (unit, functional, role-based)

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
14-09-2026 (Mon)	2	Completed	Designed and executed comprehensive unit and functional test suites evaluating billing calculations, payment processing, room allocations, and patient discharges. I formulated test cases covering diverse clinical scenarios, validating that all database transactions execute reliably and persist updated billing and admission states accurately without data loss. This verified complete operational readiness and high system maintainability.
15-09-2026 (Tue)	1.5	Completed	Conducted rigorous testing on billing calculation engines, verifying tax additions, discount deductions, and multi-payment balance adjustments. I evaluated diverse sample invoices with fractional amounts, confirming that BigDecimal math executes without precision loss or rounding anomalies across all financial reporting views. This verified comprehensive system stability, architectural consistency, and error-free operation across modules.
16-09-2026 (Wed)	1.5	Completed	Tested inpatient room allocation, transfer, and discharge state machines under concurrent user scenarios. I verified that admitting patients marks beds occupied immediately, room transfers update bed assignments atomically, and discharges calculate cumulative tariffs accurately while releasing rooms back to available status. This verified comprehensive system stability, architectural consistency, and error-free operation across modules.
17-09-2026 (Thu)	2	Completed	Validated administrative report totals against underlying MySQL database records. I executed manual verification queries calculating total revenue and admission counts, comparing them against figures displayed on admin reports.jsp. I confirmed that report aggregations match database truths precisely across all date filters. This verified comprehensive system stability, architectural consistency, and error-free operation across modules.
18-09-2026 (Fri)	1.5	Completed	Logged and documented all software defects uncovered during testing in our centralized bug tracking log. I collaborated with team members to resolve high-priority issues, including fixing a payment status filter glitch and correcting print styles on invoice.jsp. We retested all modified components to ensure complete regression stability. This guaranteed seamless clinical workflow operational efficiency.

Week 12 - Deployment (Docker, Render, cloud MySQL) and bug fixing

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
21-09-2026 (Mon)	1.75	Completed	Performed live deployment testing of the billing, admission, and reporting subsystem on the Render cloud hosting platform. I accessed the live public web application, created test invoices, allocated hospital beds, and verified that containerized Tomcat instances execute servlets and render dynamic JSP views reliably without connection timeouts or runtime exceptions.
22-09-2026 (Tue)	1.75	Completed	Investigated and resolved a decimal currency formatting anomaly observed on the cloud MySQL database server. The remote database driver parsed certain decimal fields into scientific notation strings under specific locale configurations. I enforced explicit DecimalFormat patterns in BillingDAO, ensuring standard currency formatting on all live screens. This established robust data consistency across hospital modules.
23-09-2026 (Wed)	2	Completed	Corrected a missing room availability default status in production seed scripts on the cloud database. Several newly seeded ward rooms defaulted to null availability flags, causing them to disappear from receptionist booking dropdowns. I executed an SQL update setting default flags to true, restoring full room visibility. This guaranteed seamless clinical workflow operational efficiency.
24-09-2026 (Thu)	2	Completed	Retested the complete invoice generation and printing workflow directly on the live cloud production environment. I generated invoices for discharged test patients, completed simulated payments, and verified that printable invoice layouts render cleanly without formatting shifts on Chrome, Firefox, and Safari browsers. This verified comprehensive system stability, architectural consistency, and error-free operation across modules.
25-09-2026 (Fri)	1.75	Completed	Compiled detailed deployment notes documenting cloud configuration settings, seed script updates, and decimal formatting parameters. I updated the deployment chapter of our team project report, cataloging environmental differences between local and cloud tiers. I verified that all Module 5 features operate reliably on the live public URL. This guaranteed seamless clinical workflow operational efficiency.

Week 13 - Report, PPT, demo video and viva preparation

Date (DD-MM-YYYY)	Hours	Status	Work Description (50–60 words)
28-09-2026 (Mon)	1.75	Completed	Authored the complete Billing, Admission, Reports, and Feedback chapter of our team's final Project Based Learning technical report. I documented module architecture, domain entity relationships, financial calculation algorithms, and servlet dispatching workflows. I explained how our design ensures transaction consistency, financial accuracy, and seamless administrative oversight throughout hospital operations. This verified complete system reliability.
29-09-2026 (Tue)	2	Completed	Captured high-resolution user interface screenshots and authored descriptive table walkthroughs for inclusion in the final report. I illustrated printable invoice layouts, room allocation grids, receptionist billing consoles, and administrative revenue charts. I added descriptive captions and aligned figure references with report narrative to present a clear, scholarly deliverable. This ensured dependable hospital system stability.
30-09-2026 (Wed)	2	Completed	Designed professional, high-impact presentation slides summarizing Module 5 capabilities for our upcoming academic project defense. I structured slide decks covering financial problem statements, room allocation state machines, database entity relationships, and live system screenshots. I rehearsed presentation timing, ensuring our operational highlights fit smoothly within the allotted faculty evaluation window.
01-10-2026 (Thu)	1.75	Completed	Rehearsed live system demonstration workflows showcasing invoice generation, multi-mode payment processing, room bed allocation, and patient discharge settlement. I practiced seamless handoffs between front-desk cashier actions and administrative reporting reviews, timing the presentation to fit within the allotted evaluation window. I verified that all demonstration accounts functioned reliably on the cloud server.
02-10-2026 (Fri)	1.75	Completed	Completed rigorous viva voce preparation focusing on core Java enterprise concepts, JDBC transaction management, and financial calculation patterns. I rehearsed technical explanations regarding BigDecimal precision, ACID compliance during invoice generation, and room concurrency locking. I organized project source files, ensuring complete readiness for final evaluation during the viva defense. This verified complete system reliability.

Total hours logged for Devkinandan Kumawat: 116.75 hours over 65 days.